

Learning Fluid Flows With Evolving Gaussian Processes

Second-order partial differential equations are ubiquitous in practical science and engineering, from mechanics to transport phenomena and electromagnetics. In our view, the Navier–Stokes equations governing fluid dynamics represent most, if not all, of the overall complexity of modeling these, as 1) they exhibit hybrid system behavior, such as elliptic hyperbolic, and 2) their nonlinearity results in the complex spatiotemporal dynamics that are prevalent in many practical situations. The evolving Gaussian process (E-GP) method is demonstrated on computational fluid dynamics (CFD) data describing a flow over a bluff body (a cylinder) across a range of Reynolds numbers from 100 to 1000 (the Reynolds number is a dimensionless flow rate). This deterministic, high-dimensional spatiotemporal dynamical system is well studied in fluid dynamics literature, both experimentally and numerically [S25]–[S27].

The conventional wisdom would dictate learning a separate model for each Reynolds number. However, results show that the E-GP method is capable of learning the dynamics of all flow patterns at once. Using the learned dynamics across weights of successive kernel models, the E-GP is capable of predicting the future states of functional evolution in a recursive manner. The key advantage of the E-GP is that the evolution of large function spaces can be transformed into learning the evolution in a relatively smaller Hilbert space encoded by the kernels and associated weight vector.

The CFD simulation used a fourth-order polynomial expansion with the spectral element method on the incompressible Navier–Stokes equation to generate the cylinder flow data for $Re = 100, 300, 600, 800$, and 1000 . The spatial domain is $[-2, 10] \times [-3, 3]$, excluding the diameter 1 cylinder at the origin. Neumann boundary conditions are applied to the far field of the cylinder in the y direction and the outlet of the flow field. A Dirichlet boundary condition is applied to the inlet. Each data set contains at least 200 snapshots with a uniform time step of 0.03 s. Each snapshot contains 24,000 velocity data points for $Re = 100$ or 95,000 velocity data points for $Re = 300, 600, 800$, and 1000 .

Each data set took at least 10 h in a high-performance computer cluster to generate. Figures S6 and S7(a)–(d) visualize the horizontal velocity for $Re = 100$ and 1000 , with red being the greatest negative velocity and blue the greatest positive velocity. The flow is unstable, periodic, and clearly nonlinear.

The Gaussian radial basis function kernel $k(x, y) = e^{-\|x - y\|^2 / 2\sigma^2}$ was used in the E-GP model, with σ estimated to be 0.4 . Using a budget of 600 kernel centers [see Figure S8(a)–(b) and note how the kernel centers cluster in the most dynamic regions], a 600×600 matrix $\hat{A}$ was determined, which accurately [Figure S9(a)] captured the dynamics of the nonlinear system. This was used to propagate a single initial condition w_0 forward to make predictions and then to compare the predictions to the original training data. Total percentage errors between 3% for $Re = 100$ and 7–8% for $Re = 1000$ were

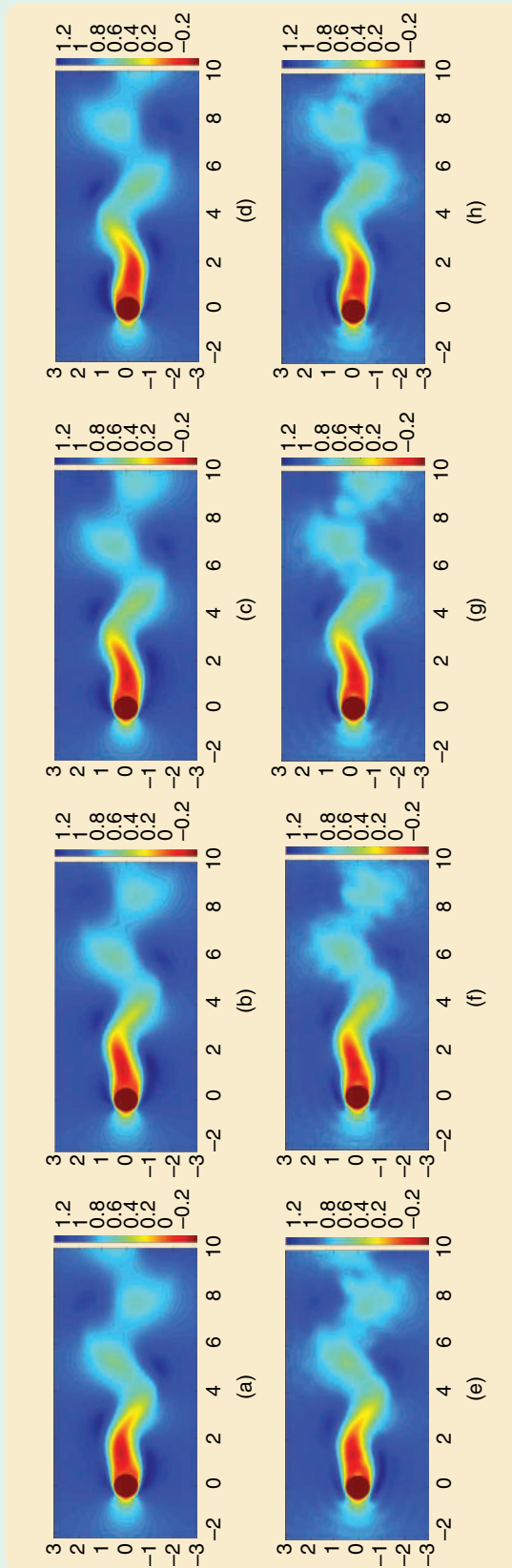


FIGURE S6 A visualization of fluid flow at $Re = 100$. (a)–(d): The computational fluid dynamics. (e)–(h): The evolving Gaussian process. (a) Snapshot 0. (b) Snapshot 10. (c) Snapshot 20. (d) Snapshot 30. (e) Snapshot 0. (f) Snapshot 10. (g) Snapshot 20. (h) Snapshot 30.

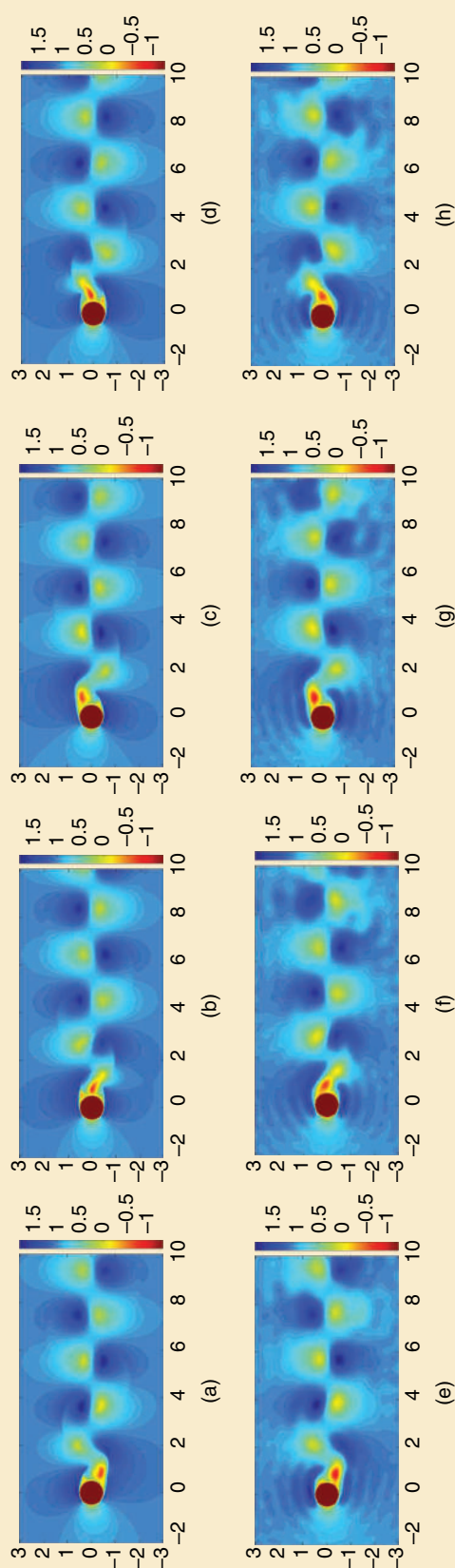


FIGURE S7 A visualization of fluid flow at $Re = 1000$. (a)–(d): The computational fluid dynamics. (e)–(h): The evolving Gaussian process. (a) Snapshot 0. (b) Snapshot 5. (c) Snapshot 10. (d) Snapshot 15. (e) Snapshot 0. (f) Snapshot 5. (g) Snapshot 10. (h) Snapshot 15.

found, as shown by the solid lines in Figure S9(a). The total percentage errors are defined as $E_\tau = (\|y_\tau - \hat{y}_\tau\|_2 / \|\hat{y}_\tau\|_2)$, where $\hat{y}_\tau$ is the output vector for time τ and y_τ is the E-GP estimate at that time. Note that *the size of the model has been reduced by almost two orders of magnitude* from the original CFD data. This process takes approximately 13 min in Matlab for a 200-snapshot $\times$ 95,000-point set on an ordinary Intel i7 4-GHz processor.

ONE TRANSITION MATRIX FOR EVERYTHING

To approach the challenge of generalizing across similar spatiotemporally evolving systems, the first question to answer is whether an $\hat{A}$ matrix can be found that accurately captures the dynamics of multiple similar flows. The answer to that question is yes, using the trajectory concatenation method. Amazingly, a single model generated this way works almost as well on all five data sets as five individual models trained separately on each data set. This is confirmed by both the total error plots [Figure S9(a)], which show only slight increases in each of the total percentage error plots, and a visual inspection of the displayed dynamic modes. This result is even more surprising given that the rate of vortex shedding for each Reynolds number is different. By taking a Fourier transform of the time evolution of a data point located at (0.5,8), it is determined that for the original data sets, the vortex shedding frequency is 0.448, 1.26, 1.38, 1.388, and 1.401 Hz for $Re = 100, 300, 600, 800$, and 1000, respectively. For the E-GP models, the frequencies are 0.452, 1.21, 1.36, 1.36, and 1.36 Hz, respectively.

GENERALIZING FROM LEARNED DYNAMICS TO UNKNOWN DYNAMICS

Having seen that it is possible to find a single transition in the weight space that models the dynamics systems across a range of parameters, the next challenge is to represent flows with parameters that the model has not been trained on. An $\hat{A}$ matrix was derived from the $Re = 100, 300, 600$, and 1000 data sets and tested against the $Re = 800$ data set. The results appear in Figure S9(b). For the first 120 snapshots, the total percentage error remains under 10%, which is satisfactory. After this, however, the total percentage error curves upward as the slight errors in the transition matrix compound. Across 800 snapshots, an average total percentage error of less than 25% was determined.

LINEAR DYNAMICAL LAYER ANALYSIS AND INSIGHTS

Due to the spatial encoding of the weights that the linear transition model operates on, we are able to analyze the dynamics and find physical insights into the process. Two techniques are demonstrated. The first uses eigendecomposition of the transition matrix to discover the eigenfunctions and invariant subspaces of the system. The second visualizes the most significant spatial interactions in the system. By marking which kernel centers are associated with different invariant

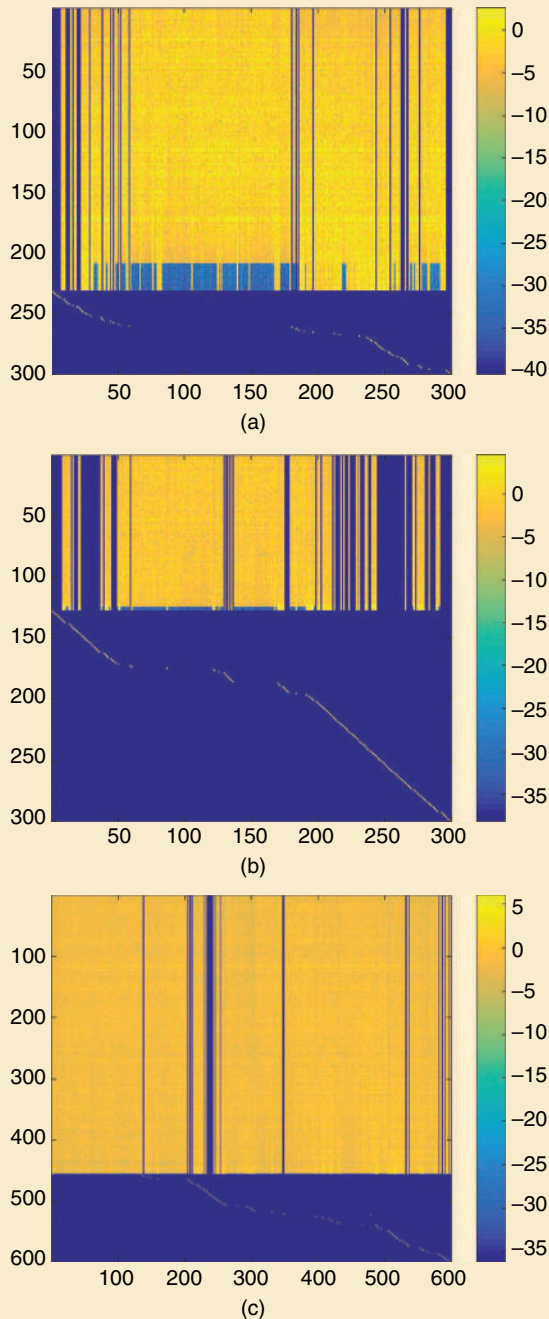


FIGURE S8 Eigenvector heat maps. (a) $Re = 100$, $\varepsilon = 0.005$. (b) $Re = 1000$, $\varepsilon = 0.05$. (c) All Reynolds numbers, $\varepsilon = 0.069$.

subspaces, the area can be spatially separated into multiple dynamic modules. The physical insight is that some regions of the space are dynamically entangled with one another, and others are independent. For those interested in monitoring spatiotemporally evolving systems, the number and loca-

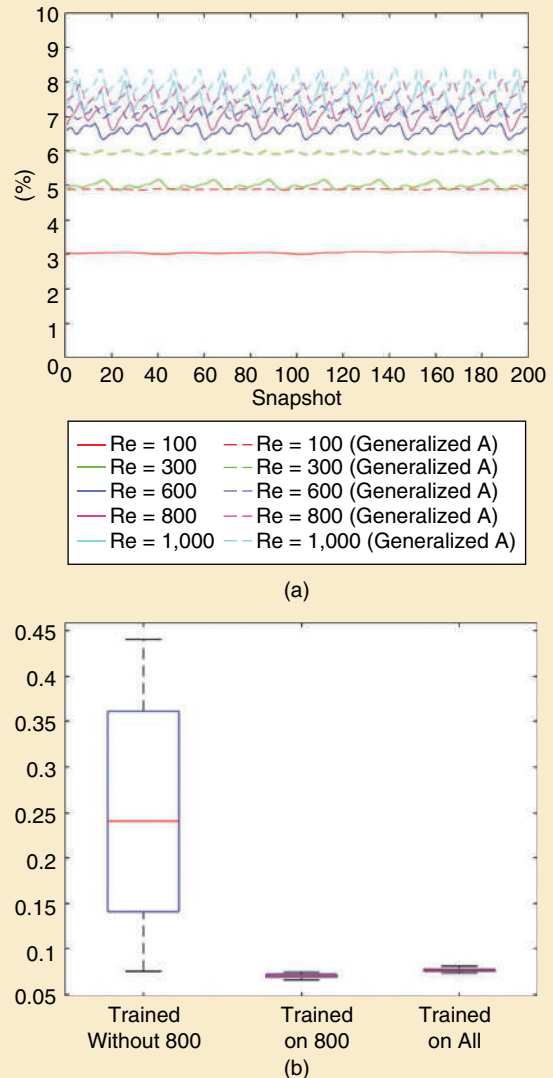


FIGURE S9 Total percentage errors. (a) The universal generalizer versus individual models. (b) Different models tested on $Re = 800$.

tion of the invariant subspaces determines how many and where feedback sensors must be for robust prediction of the system state.

Before attempting a Jordan decomposition of $\hat{A}$, any elements smaller than some small ε are zeroed to stabilize the algorithm for matrices that have many elements close to zero. Afterward, the eigenvector matrix is visualized using a *logarithmic* color chart, as seen in Figure S8(a)–(c). These plots are for models individually trained on $Re = 100$ and 1000 with 300 kernels; in addition, all five models were trained with 600 kernels, for comparison. Three classes of eigenvector are observed in the rows, constituted by the following:

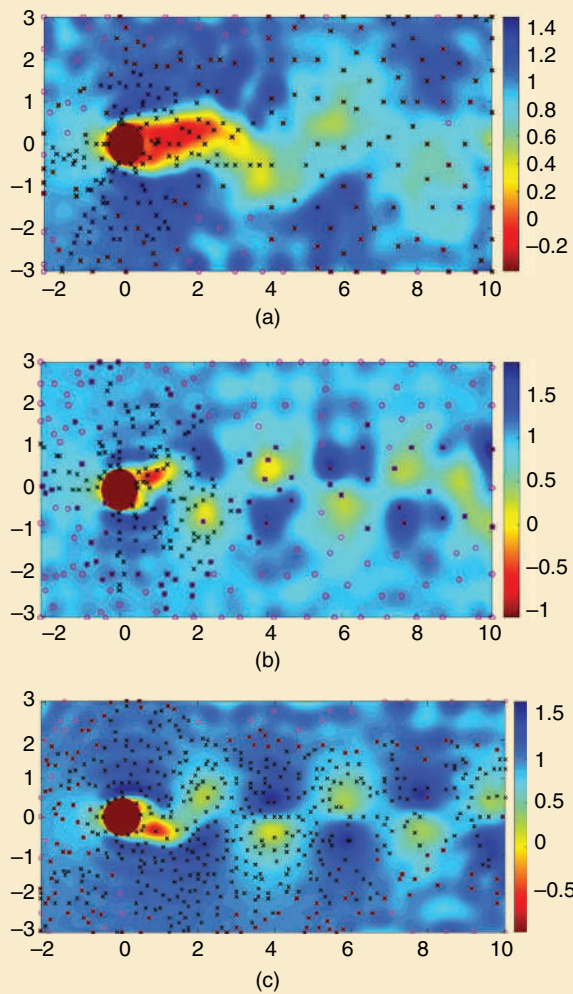


FIGURE S10 Invariant subspaces. (a) $Re = 100$, $\varepsilon = 0.005$. (b) $Re = 1000$, $\varepsilon = 0.05$. (c) All Reynolds numbers, $\varepsilon = 0.069$.

- 1) *category 1*: rows at the bottom that have exactly one nonzero element
- 2) *category 2*: a few middle rows with a dozen significant elements
- 3) *category 3*: a number of top rows that affect the majority of the kernel centers in the space.

Each eigenvector of category 1 spans its own invariant subspace and is depicted by magenta circles in Figure S10. Category 3 is one invariant subspace, depicted by black crosses. Category 2 is subsumed in category 3. The figure shows that the dynamics near/around the cylinder and in its wake are so entangled that a single sensor measurement in that area may be sufficient to estimate across that entire subspace. Alternately, areas far from the core of the dynamic excitement are their own independent, invariant subspaces and thus must be locally monitored. Another way to visualize the

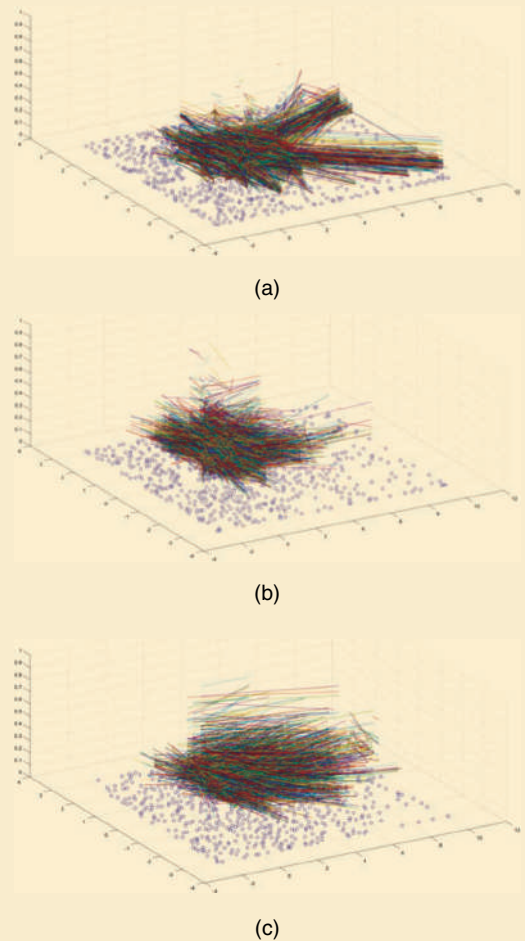


FIGURE S11 A visualization of correlations in a transition matrix. (a) $Re = 100$. (b) $Re = 1000$. (c) Trained on all five data sets.

operation of the linear transition matrix is to plot lines between kernel centers that are strongly influencing one another. That is, if a line center c_j to c_i is drawn for each of the (relatively) largest elements a_{ij} of $\hat{A}$, it can be shown how the system dynamics are coupled spatially [Figure S11(a)–(c)]. The magnitude of a_{ij} can also be plotted on a third axis for further insight into the most dominant dynamic connection in the system.

REFERENCES

- [S25] A. Roshko, "On the development of turbulent wakes from vortex streets," California Inst. of Technol., Pasadena, Rep. 1191, 1954.
- [S26] M. Braza, P. H. H. M. Chassaing, and H. H. Minh, "Numerical study and physical analysis of the pressure and velocity fields in the near wake of a circular cylinder," *J. Fluid Mech.*, vol. 165, no. 1, pp. 79–130, 1986. doi: 10.1017/S0022112086003014.
- [S27] B. N. Rajani, A. Kandasamy, and S. Majumdar, "Numerical simulation of laminar flow past a circular cylinder," *Appl. Math. Model.*, vol. 33, no. 3, pp. 1228–1247, 2009. doi: 10.1016/j.apm.2008.01.017.

Along similar lines, [37] extends the input space by adding latent variables, which enables the model to capture non-stationarity in the original space. Both these approaches require MCMC sampling for inference and thus are subject to the limitations mentioned in the preceding paragraph. A geostatistics approach that finds dynamical transition models on the linear combination of weights of a parameterized model [4], [16] is advantageous when the spatial and temporal dynamics are hierarchically separated, leading to a convex learning problem. This technique has been utilized in magnetic resonance imaging [38], [39]. As a result, complex nonstationary kernels are often not necessary (although they can be accommodated). This method is essentially the starting point of this tutorial.

A systems-theoretic study of this viewpoint enables our fundamental contributions, which are 1) enabling inference in more general domains with a larger class of basis functions than those typically considered in the geostatistics community and 2) quantifying the minimum number of measurements required to estimate the state of the system. Kalman filtering in the context of GP and kernel models has also been quite widely studied [27], [40]–[43]. There is a direct link between the Bayesian approach to inference taken in GPs and its natural extension to Kalman filters. The contributions here are in creating explicit connections between feedback observers and inference by deriving conditions for observability in the kernel space. This leads to explicit conditions for the number of sensors required and where to place them.

Sensor placement optimization is also a well-studied area. Examples include, but are not limited to, 1) geometric approaches (which seek to cover the operating space without making assumptions about the spatiotemporal dynamics [44]) and 2) information-theoretic techniques (which place their focus on sensor placement-optimizing strategies based on mutual information and information entropy for GP models [45]). It should be noted that the contribution of this article concerning sensor placement is to provide *sufficient conditions* for monitoring, rather than optimizing the placement locations, and therefore a comparison with these approaches is not considered in the experiments.

This article is connected to the large body of literature produced during the past decade about Koopman operator theory and dynamic mode decomposition (DMD), particularly in the CFD community. These methods rely on discovering *modes* of motion, which show the spatial distribution, oscillation frequency, and growth rate/decay of the component dynamics of the system. Many applications have been realized through these methods, including the ability to transform the state space so the dynamics appear linear, predicting the temporal evolution of the linear system, reconstructing the state of the original nonlinear system, and even implementing controller design. DMD is the most widely used method for finding a

finite-dimensional subspace of the Koopman operator's infinite-dimensional domain to work in [46]. Williams et al. recently integrated DMD with the kernel trick, enabling the algorithm to be extended to systems with much larger dimensions [47]. Brunton et al., inspired by DMD, were able to generate governing equations from data through the sparse identification of nonlinear dynamical systems [48]. However, these methods are restricted to approximating the Koopman operator (given a fixed vector-valued observable), and they have no way of effectively using measurements that vary in both number and location through time. Furthermore, the state of research into data-driven generalizing across similar systems with varying parameters is, at best, preliminary.

KERNEL OBSERVERS

This section outlines the modeling framework and presents theoretical results associated with the number of sampling locations required for monitoring functional evolution.

Problem Formulation

This work focuses on the predictive inference of a time-varying stochastic process whose mean f evolves temporally via $f_{\tau+1} \sim \mathbb{F}(f_{\tau}, \eta_{\tau})$, where $\mathbb{F}$ is a distribution varying with time τ and exogenous inputs η . The approach builds on the fact that, in several cases, temporal evolution can be hierarchically separated from spatial functional evolution. A classical and quite general example of this is the *abstract evolution equation* (AEO), which can be defined as the evolution of a function u embedded in a Banach space $\mathcal{B}$: $\dot{u}(t) = \mathcal{L}u(t)$, subject to $u(0) = u_0$, and where $\mathcal{L} : \mathcal{B} \rightarrow \mathcal{B}$ determines spatiotemporal transitions of $u \in \mathcal{B}$ [49]. This model of spatiotemporal evolution is very general (AEOs, for example, model many partial differential equations). However, working in Banach spaces can be computationally taxing. A simple way to make the approach computationally feasible is to place restrictions on $\mathcal{B}$: specifically, restrict the sequence f_{τ} to lie in an RKHS, the theory of which provides powerful tools for generating flexible classes of functions with relative ease [5].

In a kernel-based model, $k : \Omega \times \Omega \rightarrow \mathbb{R}$ is a positive-definite Mercer kernel on a domain Ω that represents the covariance between any two points in the input space and implies the existence of a smooth map $\psi : \Omega \rightarrow \mathcal{H}$, where $\mathcal{H}$ is an RKHS with the property $k(x, y) = \langle \psi(x), \psi(y) \rangle_{\mathcal{H}}$. The key insight behind the proposed model is that spatiotemporal evolution in the input domain corresponds to the temporal evolution of the mixing weights of a kernel model that is alone in the functional domain. Therefore, f_{τ} can be modeled by tracing the evolution of its mean embedded in an RKHS by using ODEs when the evolution is continuous or switched difference equations when it is discrete (Figure 3). The advantage of this method is that it enables the utilization of powerful ideas from systems theory for deriving necessary and sufficient conditions for spatiotemporal

monitoring. This article focuses on the class of functional evolutions $\mathbb{F}$ defined by linear Markovian transitions in an RKHS. While extension to the nonlinear case is possible (and nontrivial), it is not pursued here so as to ease the exposition of the key ideas. The class of linear transitions

in an RKHS is rich enough to approximately model many real-world data sets, as suggested by the experiments.

Let $y \in \mathbb{R}^N$ be the measurements of the function available from N sensors, $\mathcal{A}: \mathcal{H} \rightarrow \mathcal{H}$ be a linear transition operator in the RKHS $\mathcal{H}$, and $\mathcal{K}: \mathcal{H} \rightarrow \mathbb{R}^N$ be a linear measurement operator. The model for the functional evolution and measurement studied in this article is

$$f_{\tau+1} = \mathcal{A}f_{\tau} + \eta_{\tau}, \quad y_{\tau} = \mathcal{K}_{\tau}f_{\tau} + \zeta_{\tau}, \quad (1)$$

where η_{τ} is a zero-mean stochastic process in $\mathcal{H}$, and ζ_{τ} is a Wiener process in $\mathbb{R}^N$. Classical treatments of kernel methods emphasize that, for most kernels, the feature map ψ is unknown and possibly infinite dimensional. This forces practitioners to work in the dual space of $\mathcal{H}$, whose dimensionality is the number of samples in the data set being modeled. This conventional wisdom precludes the use of kernel methods for most tasks involving modern data sets, which may have millions (and sometimes billions) of samples [50]. An alternative is to work with a feature map $\hat{\psi}(x) := [\hat{\psi}_1(x) \dots \hat{\psi}_M(x)]^T$ to an approximate feature space $\hat{\mathcal{H}}$ with the property that for every element $f \in \mathcal{H}$, $\exists \hat{f} \in \hat{\mathcal{H}}$, subject to $\|f - \hat{f}\| < \epsilon$, for an appropriate function norm, and $\epsilon > 0$. A few such approximations are listed as follows.

Dictionary of Atoms

Let Ω be compact. Given points $C = \{c_1, \dots, c_M\}$, $c_i \in \Omega$, this results in a dictionary of atoms (Figure 4) $\mathcal{F}^C = \{\psi(c_1), \dots, \psi(c_M)\}$, $\psi(c_i) \in \mathcal{H}$, the span of which is a strict subspace $\hat{\mathcal{H}}$ of the RKHS $\mathcal{H}$ generated by the kernel. Here,

$$\hat{\psi}_i(x) := \langle \psi(x), \psi(c_i) \rangle_{\mathcal{H}} = k(x, c_i). \quad (2)$$

Low-Rank Approximations

Let Ω be compact, let $C = \{c_1, \dots, c_M\}$, $c_i \in \Omega$, and let $K \in \mathbb{R}^{M \times M}$, $K_{ij} := k(c_i, c_j)$ be the Gram matrix computed from C . This matrix can be diagonalized to compute approximations $[\hat{\lambda}_i, \hat{\phi}_i(x)]$ of the eigenvalues and eigenfunctions $[\lambda_i, \phi_i(x)]$ of the kernel [51]. These spectral quantities can then be used to compute $\hat{\psi}_i(x) := \sqrt{\hat{\lambda}_i} \hat{\phi}_i(x)$.

Random Fourier Features

Let $\Omega \subset \mathbb{R}^n$ be compact, and let $k(x, y) = e^{-\|x-y\|^2/2\sigma^2}$ be the Gaussian radial basis function (RBF) kernel. Then, random Fourier features approximate the kernel feature map as $\hat{\psi}_{\omega}: \Omega \rightarrow \hat{\mathcal{H}}$, where ω is a sample from the Fourier transform of $k(x, y)$, with the property that $k(x, y) = \mathbb{E}_{\omega}[\langle \hat{\psi}_{\omega}(x), \hat{\psi}_{\omega}(y) \rangle_{\hat{\mathcal{H}}}]$ [50]. In this case, if $V \in \mathbb{R}^{M/2 \times n}$ is a random matrix representing the sample ω , then $\hat{\psi}_i(x) := [(1/\sqrt{M})\sin([Vx]_i), (1/\sqrt{M})\cos([Vx]_i)]$. Similar approximations exist for other radially symmetric kernels as well as dot product kernels.

In the approximate space case, replace the transition operator $\mathcal{A}: \mathcal{H} \rightarrow \mathcal{H}$ in (1) with $\hat{\mathcal{A}}: \hat{\mathcal{H}} \rightarrow \hat{\mathcal{H}}$. This

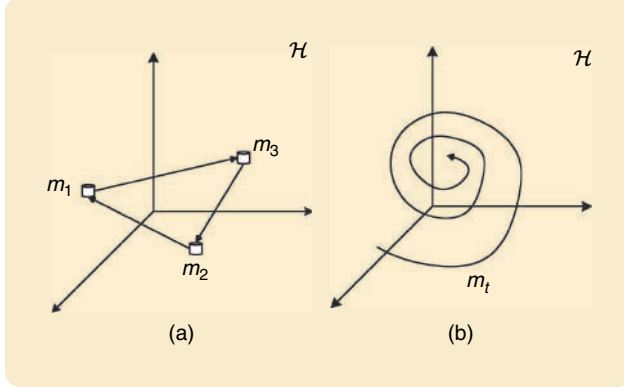


FIGURE 3 Two types of Hilbert space evolutions. (a) Discrete switches in reproducing kernel Hilbert space $\mathcal{H}$. (b) Smooth evolution in $\mathcal{H}$.

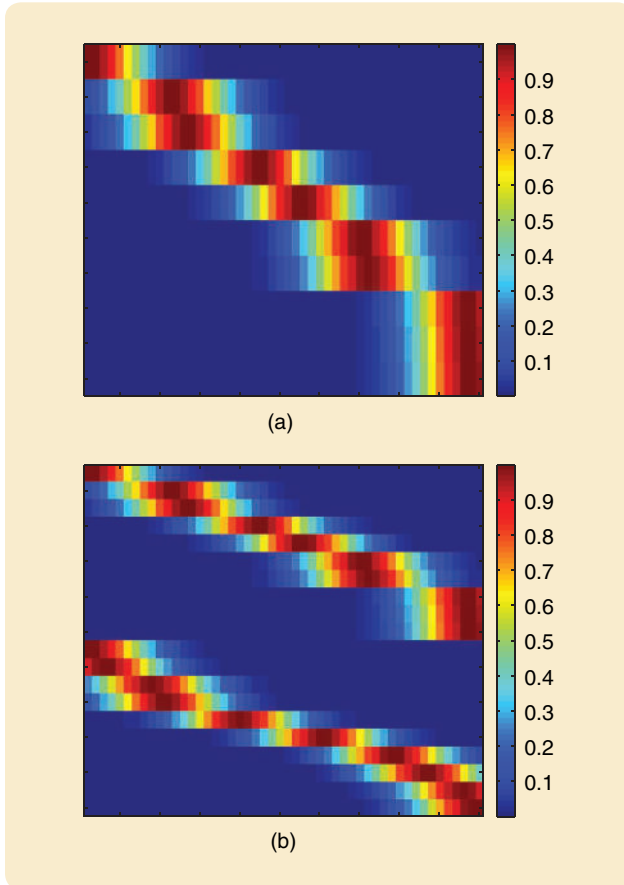


FIGURE 4 Shaded observation matrices for a dictionary of atoms. Each row represents a sensing location, with the color map indicating the evaluation of the kernel function with respect to the others points in the domain. (a) One shaded (Definition 1). (b) Two shaded [(8)].

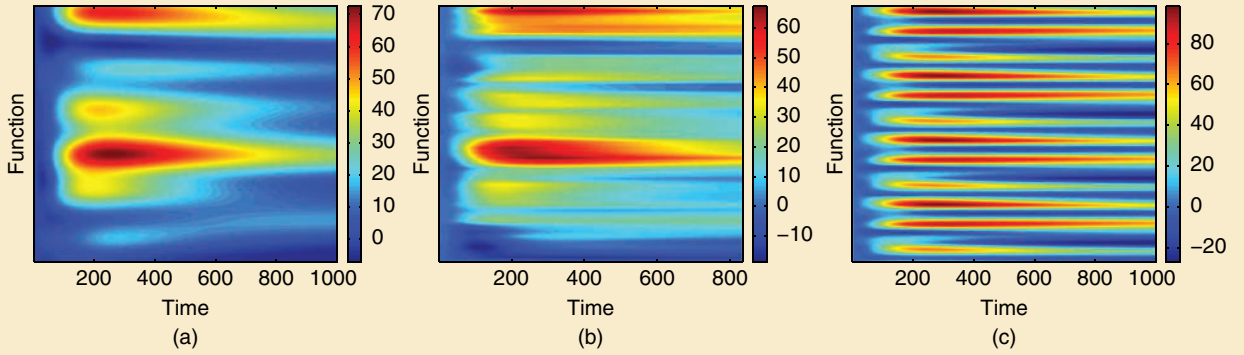


FIGURE 5 The 1D function evolution across a fixed transition matrix A , initial condition w_0 , and centers C [but with different kernels $k(x, y)$]. Each y -vector at a given value of x represents the output of the function, which evolves from left to right. As seen, changing the kernel creates very different dynamic behaviors. (a) Gaussian. (b) Laplacian. (c) Periodic.

approximate regime, which combines the flexibility of a truly nonparametric approach with computational realizability, still enables the representation of rich phenomena (as will be seen in the sequel and Figure 5).

The finite-dimensional evolution equations approximating (1) in dual form (Figure 6) are

$$w_{\tau+1} = \hat{A}w_{\tau} + \eta_{\tau}, \quad y_{\tau} = Kw_{\tau} + \zeta_{\tau}, \quad (3)$$

where $\hat{A} \in \mathbb{R}^{M \times M}$, $K \in \mathbb{R}^{N \times M}$; the vectors $w_{\tau} \in \mathbb{R}^M$; and a slightly abused notation enables y_{τ}, η_{τ} and ζ_{τ} denote their counterparts in $\hat{\mathcal{H}}$. Here, K is the matrix whose rows are of the form $K_{(i)} = \hat{\Psi}(x_i) = [\hat{\psi}_1(x_i) \ \hat{\psi}_2(x_i) \ \cdots \ \hat{\psi}_M(x_i)]$. In systems theoretic language, each row of K corresponds to a *measurement* at a specific location, and the matrix itself acts as a measurement operator. The equations in (1) suggest an immediate extension to functional control problems. Select another basis for $\mathcal{H}$ as $\tilde{\psi}(x) := [\tilde{\psi}_1(x) \ \cdots \ \tilde{\psi}_{\ell}(x)]^T$, where the functions $\tilde{\psi}_j(x)$ are used to approximate the RKHS $\mathcal{H}$ generated by the kernel. Denote the span of these functions as $\tilde{\mathcal{H}}$. In the dictionary of atoms case, an example would be another set of atoms $\mathcal{F}_D = [\psi(d_1) \ \cdots \ \psi(d_{\ell})]$, $\psi(d_j) \in \mathcal{H}, d_j \in \Omega$, with $\tilde{\mathcal{H}}$ being a strict subspace of the RKHS $\mathcal{H}$ generated by the kernel. The functional evolution equations are then

$$f_{\tau+1} = \mathcal{A}f_{\tau} + \mathcal{B}\delta_{\tau} + \eta_{\tau}, \quad y_{\tau} = \mathcal{K}f_{\tau} + \zeta_{\tau}, \quad (4)$$

where the control functions δ_{τ} evolve in $\tilde{\mathcal{H}}$ and $\mathcal{B}: \tilde{\mathcal{H}} \rightarrow \hat{\mathcal{H}}$.

To derive the finite-dimensional equivalent of $\mathcal{B}$, the structure of the matrix B must be determined. Since $\hat{\mathcal{H}}$ is not, in general, isomorphic to $\tilde{\mathcal{H}}$, this imposes strict restrictions on B . It is possible to derive B using least squares, employing the inner product of $\mathcal{H}$. An instructive example is where both $\hat{\mathcal{H}}$ and $\tilde{\mathcal{H}}$ are generated by dictionaries of atoms. Recall that, in this case, $\mathcal{F}^C = [\psi(c_1) \ \cdots \ \psi(c_M)]$ is the basis for $\hat{\mathcal{H}}$, $\delta = \sum_{j=1}^{\ell} \hat{w}_j \psi(d_j)$, and $\mathcal{F}^C = [\psi(c_1) \ \cdots \ \psi(c_M)]$

is the basis for $\mathcal{H}^C$. The projection of δ onto $\hat{\mathcal{H}}$ can be derived as

$$\begin{bmatrix} \langle \delta, \psi(c_1) \rangle_{\mathcal{H}} \\ \vdots \\ \langle \delta, \psi(c_M) \rangle_{\mathcal{H}} \end{bmatrix} = \underbrace{\begin{bmatrix} \langle \psi(d_1), \psi(c_1) \rangle_{\mathcal{H}} & \cdots & \langle \psi(d_{\ell}), \psi(c_1) \rangle_{\mathcal{H}} \\ \vdots & \ddots & \vdots \\ \langle \psi(d_1), \psi(c_M) \rangle_{\mathcal{H}} & \cdots & \langle \psi(d_{\ell}), \psi(c_M) \rangle_{\mathcal{H}} \end{bmatrix}}_{K_{CD}} \begin{bmatrix} \hat{w}_1 \\ \vdots \\ \hat{w}_{\ell} \end{bmatrix}. \quad (5)$$

Note that in the dictionary of atoms case, the entries of K_{CD} can be computed in closed form as $K_{CDij} := k(d_i, c_j)$ using the reproducing property. This derivation shows that the operator B is simply $K_{CD} \in \mathbb{R}^{M \times \ell}$, the kernel matrix between the data C generating the atoms $\mathcal{F}^C$ of $\hat{\mathcal{H}}$ and the data D generating the atoms $\mathcal{F}_D$ of $\tilde{\mathcal{H}}$. Thus, the finite-dimensional evolution equations equivalent to (4) are

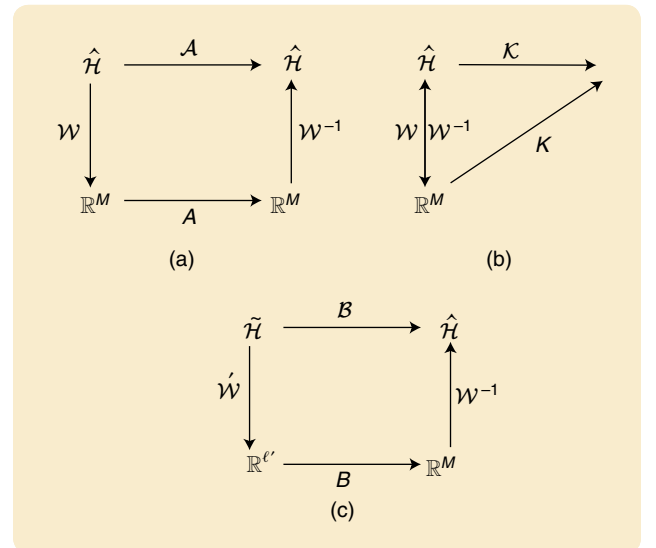


FIGURE 6 Commutative diagrams between primal and dual spaces. (a) The relationship between $\mathcal{A}$ and A . (b) The relationship between $\mathcal{K}$ and K . (c) The relationship between $\mathcal{B}$ and B .

$$w_{\tau+1} = \hat{A}w_{\tau} + K_{CD}\dot{w}_{\tau}, \quad y_{\tau} = K_{\tau}w_{\tau}. \quad (6)$$

Let the generalized observability matrix [52] be defined as

$$O_Y = \begin{bmatrix} K\hat{A}^{\tau_1} \\ \vdots \\ K\hat{A}^{\tau_L} \end{bmatrix},$$

where $Y = \{\tau_1, \dots, \tau_L\}$ are the set of instances τ_i when applying the operator K . A linear system is said to be *observable* if O_Y has full column rank [that is, $\text{Rank}(O_Y) = M$] for $Y = \{0, 1, \dots, M-1\}$ [52]. Observability guarantees two critical facts. First, the state w_0 can be exactly recovered from a finite series of measurements $\{y_{\tau_1}, y_{\tau_2}, \dots, y_{\tau_L}\}$. Specifically, defining $y_Y = [y_{\tau_1}^T, y_{\tau_2}^T, \dots, y_{\tau_L}^T]^T$, $y_Y = O_Y w_0$. Second, it guarantees that a feedback-based *observer* can be designed such that the estimate of w_{τ} , denoted by $\hat{w}_{\tau}$, converges exponentially fast to w_{τ} in the limit of the samples. Note that all theoretical results assume $\hat{A}$ is available. While system identification is performed in the experiments [21], it is not the focus of this article.

We are now in a position to formally state the spatiotemporal modeling, control, and inference problems being considered. **Given a spatiotemporally evolving system modeled using (3), choose a set of N sensing locations such that, even with $N \ll M$, the functional evolution of the spatiotemporal model can be estimated (which corresponds to *monitoring*), robustly predicted (which corresponds to *Bayesian filtering*), and controlled (which corresponds to *functional control*).** This approach to solve the monitoring and prediction problem relies on the design of the measurement operator K so that the pair $(K, \hat{A})$ is observable: any Bayesian state estimator (for example, a Kalman filter) utilizing this pair is denoted as a *kernel observer* (Figure 7). In the case where no measurements are taken, for the sake of consistency, denote the state estimator as an *autonomous* kernel observer. In the controls case, given a spatiotemporally evolving system modeled using (6), a set of N sensing locations and ℓ' control locations must be chosen such that, even with $N \ll M, \ell' \ll M$, the functional evolution of the spatiotemporal model can be controlled. In this case, both a

measurement operator K and a control operator K_{CD} must be designed such that the pair $(K_{CD}, \hat{A})$ is controllable. A controls system utilizing this pair and the measurement operator K is denoted as a *kernel controller*.

Preliminaries of Rational Canonical Structures

Using a geometric approach to the choice of sampling locations for inferring w_{τ} in (3), the extension to control is similar. Use the notation $\mathcal{V}$, with $\dim(\mathcal{V}) = M$, to emphasize the fact that these theorems hold for any finite-dimensional vector space. Consider the linear operator $\mathcal{A}: \mathcal{V} \rightarrow \mathcal{V}$, and recall that the definition of observability requires the construction of a linear operator $\mathcal{K}: \mathcal{V} \rightarrow \mathcal{U}$, with $\dim(\mathcal{U}) = N$, such that $\text{rank}[(\mathcal{K})^T \dots (\mathcal{K}\mathcal{A}^{M-1})^T]^T = M$. In most applications, if $N \geq M$ and $\text{rank}(\mathcal{K}) = N$, it is reasonable to expect that observability may be achieved. However, for this purpose, N must be *significantly* less than M . Therefore, $\mathcal{K}$ must be designed with as small a rank as possible. To do so, we require a series of vectors v_i that, under repeated iterations of $\mathcal{A}$, can generate a basis for $\mathcal{V}$. For this task, a fundamental decomposition result from the theory of modules (known as the *rational canonical structure*) of $\mathcal{A}$ [53] is used.

The intuition here is that if the sequence $\{v_i\}_i$ can generate this basis, it can be directly used to construct $\mathcal{K}$. The linear operator $\mathcal{A}: \mathcal{V} \rightarrow \mathcal{V}$ has a characteristic polynomial $\pi(\lambda)$ such that $\pi(\mathcal{A}) = 0$ by the Cayley–Hamilton theorem. The minimal polynomial (MP) of $\mathcal{A}$ is the monic polynomial $\alpha(\cdot)$ of the least degree [denoted by $\deg(\cdot)$], given as $\alpha(\lambda) = a_0 + a_1\lambda + \dots + \lambda^{\deg(\alpha)} = 0$ such that $\alpha(\mathcal{A}) = a_0I + a_1\mathcal{A} + \dots + \mathcal{A}^{\deg(\alpha)} = 0$. The MP is unique and divides $\pi(\lambda)$ so that $\deg(\alpha) \leq \deg(\pi)$. The MP of a vector $v \in \mathcal{V}$ relative to $\mathcal{A}$ is the unique monic polynomial ξ_v of the least degree such that $\xi_v(\mathcal{A})v = a_0v + a_1\mathcal{A}v + \dots + \mathcal{A}^{\deg(\alpha)}v = 0$. If $\deg(\alpha) = M$, then $\mathcal{A}$ is *cyclic*, and $\exists v \in \mathcal{V}$ such that the vectors $\{v, \mathcal{A}v, \dots, \mathcal{A}^{M-1}v\}$ form a basis for $\mathcal{V}$. This is the same as saying that the pair $(v^T, \mathcal{A}^T)$ is observable. A subspace $\mathcal{V}_S \subset \mathcal{V}$ subject to $\mathcal{A}\mathcal{V}_S \subset \mathcal{V}_S$ is $\mathcal{A}$ -cyclic if $\mathcal{A}|_{\mathcal{V}_S}$, the restriction of $\mathcal{A}$ to the subspace $\mathcal{V}_S$, is cyclic. If $\alpha(\lambda)$ is the MP of $\mathcal{A}$ and $\deg(\alpha) = m < M$, $\exists v \in \mathcal{V}$ such that $\{v, \mathcal{A}v, \dots, \mathcal{A}^{m-1}v\}$ span an m -dimensional $\mathcal{A}$ -cyclic subspace $\mathcal{V}_S$, with v being the *cyclic generator* of $\mathcal{V}_S$. The subspace $\mathcal{V}_S$ decomposes $\mathcal{V}$ relative to $\mathcal{A}$. By the rational canonical structure theorem [53, Th. 0.1], $\mathcal{A}$ can be

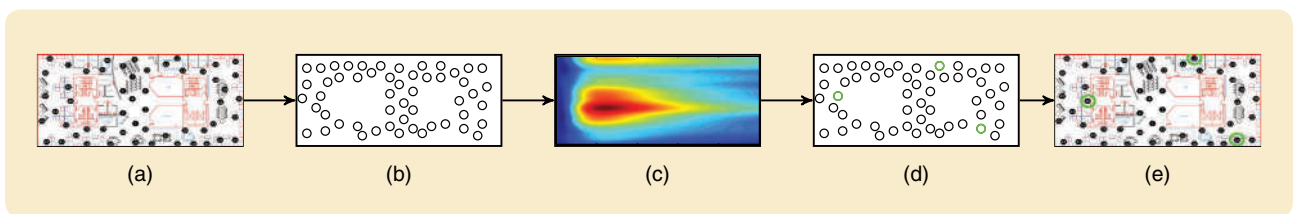


FIGURE 7 A description of how the kernel observer fits in the sensing framework. Physical locations are mapped to data locations across which historical information is collected as a time series. Functional inference is performed across $\hat{\mathcal{H}}$ to solve for $\hat{A}$. The measurement operator K is then computed (see Figure 8), leading to the sensor placement. (a) Physical sampling locations. (b) Data locations. (c) The functional inference (for $\hat{A}$). (d) Sensor location selection after basis decomposition ($\ell = 3$). (e) Physical sensor placement.

successively decomposed into subspaces $\mathcal{V}_i \subset \mathcal{V}, i \in \{1, \dots, \ell\}$, subject to $\mathcal{V} = \mathcal{V}_1 \oplus \dots \oplus \mathcal{V}_\ell$, $\mathcal{A}\mathcal{V}_i \subset \mathcal{V}_i$, and $\mathcal{A}|_{\mathcal{V}_i}, i \in \{1, \dots, \ell\}$ are cyclic.

In general, the subspaces $\mathcal{V}_i$ are not unique for a fixed $\mathcal{A}$. The integer ℓ is unique and called the *cyclic index* of $\mathcal{A}$. One of our main results shows that the cyclic index is a lower bound on the number of measurements required to reconstruct w_τ (see Proposition 3 and Algorithm 1). The matrix transform associated with this theorem is known as the *Frobenius normal form* (denoted by $C \in \mathbb{R}^{M \times M}$): for $\mathcal{A} \in \mathbb{R}^{M \times M}$, $\exists Q \in \mathbb{R}^{M \times M}$ is invertible such that $\mathcal{A} = QCQ^{-1}$. We will also use *Jordan decomposition*, where for $\mathcal{A} \in \mathbb{R}^{M \times M}$, $\exists P \in \mathbb{R}^{M \times M}$ is invertible such that $\mathcal{A} = P\Lambda P^{-1}$, where Λ is a unique block diagonal matrix with Jordan blocks with λ_i along the diagonal. If all the eigenvalues λ_i are nonzero and real, we say the matrix has a *full-rank Jordan decomposition*.

MAIN RESULTS

This section proves the results concerning the observability of spatiotemporally varying functions modeled by the functional evolution and measurement (3). Specifically, observability of the system states implies that the current state of the spatiotemporally varying function can be recovered using a small number of sampling locations N , which enables 1) tracking the function and 2) predicting the function's evolution forward in time. Using the approximation $\hat{\mathcal{H}} \approx \mathcal{H}$, given M basis functions, implies that the dual space of $\hat{\mathcal{H}}$ is $\mathbb{R}^M$. Proposition 1 shows that if $\hat{A}$ has a full-rank Jordan decomposition, the observation matrix K that meets a condition called *shadedness* (Definition 1) is sufficient for the system to be observable. Proposition 2 provides a lower bound on the number of sampling locations required for observability, which holds for any $\hat{A}$. Proposition 3 constructively shows the existence of an abstract measurement map $\tilde{K}$ achieving this lower bound. Since the measurement map does not have the structure of a kernel matrix, a slightly weaker sufficient condition for the observability of any $\hat{A}$ is in Theorem 1. Finally, since both K and K_{CD} are kernel matrices generated from a shared kernel, these observability results translate directly into controllability results.

Definition 1: Shaded Observation Matrix

Given that $k: \Omega \times \Omega \rightarrow \mathbb{R}$ is positive definite on a domain Ω , let $\{\hat{\psi}_1(x), \dots, \hat{\psi}_M(x)\}$ be the set of bases generating an approximate feature map $\hat{\psi}: \Omega \rightarrow \hat{\mathcal{H}}$, and let $\mathcal{X} = \{x_1, \dots, x_N\}$ be the set of sampling (or sensing) locations, with each $x_i \in \Omega$. Let $K \in \mathbb{R}^{N \times M}$ being the observation matrix, where $K_{ij} := \hat{\psi}_j(x_i)$. For each row $K_{(i)} := [\hat{\psi}_1(x_i) \dots \hat{\psi}_M(x_i)]$, define the set $\mathcal{I}_{(i)} := \{l_1^{(i)}, l_2^{(i)}, \dots, l_{M_i}^{(i)}\}$ to be the indices in the observation matrix row i that are nonzero. Then, if $\cup_{i \in \{1, \dots, N\}} \mathcal{I}_{(i)} = \{1, 2, \dots, M\}$, denote K as a *shaded observation matrix* [see Figure 4(a)]. This definition seems quite abstract. Hence, the following remark considers a more concrete example.

Remark 1

Let $\hat{\psi}$ be generated by the dictionary given by $\mathcal{C} = \{c_1, \dots, c_M\}$, $c_i \in \Omega$. Note that since $\hat{\psi}_j(x_i) = \langle \psi(x_i), \psi(c_j) \rangle_{\mathcal{H}} = k(x_i, c_j)$, K is the kernel matrix between $\mathcal{X}$ and $\mathcal{C}$. For the kernel matrix to be shaded, the implication is that there does not exist an atom $\psi(c_j)$ such that the projections $\langle \psi(x_i), \psi(c_j) \rangle_{\mathcal{H}}$ vanish for all x_i , $1 \leq i \leq N$. Intuitively, the shadedness property requires that the sensor locations x_i are privy to information propagating from every c_j . As an example, note that (in principle) for the Gaussian kernel, a single row generates a shaded kernel matrix. However, in this case, the matrix can have many entries that are extremely close to zero and will probably be very ill conditioned.

Proposition 1

Given that $k: \Omega \times \Omega \rightarrow \mathbb{R}$ is positive definite on a domain Ω , let $\{\hat{\psi}_1(x), \dots, \hat{\psi}_M(x)\}$ be the set of bases generating an approximate feature map $\hat{\psi}: \Omega \rightarrow \hat{\mathcal{H}}$, and let $\mathcal{X} = \{x_1, \dots, x_N\}$, $x_i \in \Omega$. Consider the discrete linear system on $\hat{\mathcal{H}}$ given by the evolution and measurement (3). Suppose that a full-rank Jordan decomposition of $\hat{A} \in \mathbb{R}^{M \times M}$ of the form $\hat{A} = P\Lambda P^{-1}$ exists, where $\Lambda = [\Lambda_1 \dots \Lambda_O]$ and there are no repeated eigenvalues. Then, given a set of time instances $\Upsilon = \{\tau_1, \tau_2, \dots, \tau_L\}$ and a set of sampling locations $\mathcal{X} = \{x_1, \dots, x_N\}$, the system (3) is observable if the observation matrix K_{ij} is shaded according to Definition 1, Υ has distinct values, and $|\Upsilon| \geq M$.

Proof

Consider a system where $\hat{A} = \Lambda$, with Jordan blocks $\{\Lambda_1, \Lambda_2, \dots, \Lambda_O\}$ along the diagonal. Then, $\hat{A}^{\tau_i} = \text{diag}([\Lambda_1^{\tau_i} \ \Lambda_2^{\tau_i} \ \dots \ \Lambda_O^{\tau_i}])$. This results in

$$\mathcal{O}_\Upsilon = \begin{bmatrix} K\hat{A}^{\tau_1} \\ \dots \\ K\hat{A}^{\tau_L} \end{bmatrix}. \quad \mathcal{O}_\Upsilon \in \mathbb{R}^{NL \times M}$$

ALGORITHM 1 Measurement map $\tilde{K}$

Input: $\hat{A} \in \mathbb{R}^{M \times M}$

Compute Rational Canonical Form, subject to $C = Q^{-1}\hat{A}^T Q$.

Set $C_0 := C$ and $M_0 := M$.

for $i = 1$ to ℓ **do**

Obtain (minimum polynomial) $\alpha_i(\lambda)$ of C_{i-1} . This returns associated indices $\mathcal{J}^{(i)} \subset \{1, 2, \dots, M_{i-1}\}$.

Construct vector $v_i \in \mathbb{R}^M$ such that $\xi_{v_i}(\lambda) = \alpha_i(\lambda)$.

Use indices $\{1, 2, \dots, M_{i-1}\} \setminus \mathcal{J}^{(i)}$ to select matrix C_i . Set $M_i := |\{1, 2, \dots, M_{i-1}\} \setminus \mathcal{J}^{(i)}|$

end for

Compute $\hat{K} = [v_1^T, v_2^T, \dots, v_\ell^T]^T$

Output: $\tilde{K} = \hat{K}Q^{-1}$

It must be proved that the column rank of $\mathcal{O}_Y$ is M , which is not immediately obvious since typically $N \ll M$. To prove the statement, it is shown that computing the rank of $\mathcal{O}_Y$ is equivalent to the rank computation of the product of two simple matrices. In the following, use the notation $\mathbf{0}_{\mathbb{R}^{l \times j}}$ to denote an $l \times j$ matrix of all zeros.

In the first step, write the preceding matrix as the product of two matrices. It can be shown that $\mathcal{O}_Y$ is the product of two block matrices,

$$\mathcal{O}_Y = \underbrace{\begin{bmatrix} K & \cdots & \mathbf{0}_{\mathbb{R}^{N \times M}} \\ \vdots & \ddots & \vdots \\ \mathbf{0}_{\mathbb{R}^{N \times M}} & \cdots & K \end{bmatrix}}_{\hat{\mathbf{K}} \in \mathbb{R}^{NL \times ML}} \underbrace{\begin{bmatrix} \Lambda_1^{\tau_1} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \Lambda_O^{\tau_1} \\ \vdots & \ddots & \vdots \\ \Lambda_1^{\tau_L} & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & \Lambda_O^{\tau_L} \end{bmatrix}}_{\hat{\mathbf{A}} \in \mathbb{R}^{ML \times M}}.$$

Note that $\hat{\mathbf{K}}$ must be simplified even further. Recall that a matrix's rank is preserved under a product with an invertible matrix. Design a matrix of elementary row operations $U \in \mathbb{R}^{N \times N}$ such that $\check{\mathbf{K}} := UK$ is a matrix with at least one row vector of nonzeros; this can be achieved by having an elementary matrix that adds rows together. By the shadedness assumption, such a matrix exists. This operation can be written as

$$UK = \begin{bmatrix} \check{K}_{11} & \check{K}_{12} & \cdots & \check{K}_{1M} \\ \check{K}_{21} & \check{K}_{22} & \cdots & \check{K}_{2M} \\ \vdots & \vdots & \ddots & \vdots \\ \check{K}_{N1} & \check{K}_{N2} & \cdots & \check{K}_{NM} \end{bmatrix}.$$

Without a loss of generality (and slightly abusing the notation), let this multiplication lead to one nonzero row (with the rest of the elements of the matrix being zero) as

$$UK = \begin{bmatrix} k_{11} & k_{12} & \cdots & k_{1M} \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix}.$$

Since elementary matrices are full rank, $\text{rank}(UK) = \text{rank}(K)$.

To analyze the rank of $\mathcal{O}_Y$, apply these elementary matrices to every $K \in \hat{\mathbf{K}}$. Consider the block diagonal matrix $U \in \mathbb{R}^{NL \times NL}$, with $U \in \mathbb{R}^{N \times N}$ along the diagonal and zeros everywhere else. That is,

$$U := \begin{bmatrix} U & \mathbf{0}_{\mathbb{R}^{N \times N}} & \cdots & \mathbf{0}_{\mathbb{R}^{N \times N}} \\ \mathbf{0}_{\mathbb{R}^{N \times N}} & U & \cdots & \mathbf{0}_{\mathbb{R}^{N \times N}} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}_{\mathbb{R}^{N \times N}} & \mathbf{0}_{\mathbb{R}^{N \times N}} & \cdots & U \end{bmatrix}. \quad (7)$$

It can be shown that U is full rank; in other words, it has rank NL . Referring back to the observability matrix,

$$\mathcal{U}\mathcal{O}_Y = U\hat{\mathbf{K}}\hat{\mathbf{A}} = \underbrace{\begin{bmatrix} UK & \mathbf{0}_{\mathbb{R}^{N \times M}} & \cdots & \mathbf{0}_{\mathbb{R}^{N \times M}} \\ \mathbf{0}_{\mathbb{R}^{N \times M}} & UK & \cdots & \mathbf{0}_{\mathbb{R}^{N \times M}} \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{0}_{\mathbb{R}^{N \times M}} & \mathbf{0}_{\mathbb{R}^{N \times M}} & \cdots & UK \end{bmatrix}}_{U\hat{\mathbf{K}} \in \mathbb{R}^{NL \times ML}} \underbrace{\hat{\mathbf{A}}}_{\in \mathbb{R}^{ML \times M}},$$

since $\mathbf{0}_{\mathbb{R}^{N \times N}}\mathbf{0}_{\mathbb{R}^{N \times M}} = \mathbf{0}_{\mathbb{R}^{N \times M}}$. Due to the fact that $\text{rank}(\mathcal{U}\mathcal{O}_Y) = \text{rank}(\mathcal{O}_Y)$, the rank analysis can therefore be performed on the simpler matrix $\text{rank}(U\hat{\mathbf{K}}\hat{\mathbf{A}})$. Note that

$$\begin{aligned} U\hat{\mathbf{K}}\hat{\mathbf{A}}^{\tau_j} &= \begin{bmatrix} k_{11} & k_{12} & \cdots & k_{1M} \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{bmatrix} \hat{\mathbf{A}}^{\tau_j} \\ &= \begin{bmatrix} k_{11}\lambda_1^{\tau_j} \begin{pmatrix} \tau_j \\ 1 \end{pmatrix} \lambda_1^{\tau_j-1} + k_{12}\lambda_1^{\tau_j} & \cdots & k_{1M}\lambda_O^{\tau_j} \\ 0 & & 0 \\ \vdots & & \vdots \\ 0 & & 0 \end{bmatrix}. \end{aligned}$$

Thus, following some more elementary row operations encoded by $V \in \mathbb{R}^{ML \times ML}$,

$$\begin{aligned} VU\mathcal{O}_Y &= \begin{bmatrix} k_{11}\lambda_1^{\tau_1} & \cdots & k_{1M}\lambda_O^{\tau_1} \\ k_{11}\lambda_1^{\tau_2} & \cdots & k_{1M}\lambda_O^{\tau_2} \\ \vdots & \ddots & 0 \\ k_{11}\lambda_1^{\tau_L} & \cdots & k_{1M}\lambda_O^{\tau_L} \\ \mathbf{0}_{\mathbb{R}^{M(L-1) \times 1}} & \cdots & \mathbf{0}_{\mathbb{R}^{M(L-1) \times 1}} \end{bmatrix} \\ &= \begin{bmatrix} \Phi \\ \mathbf{0}_{\mathbb{R}^{M(L-1) \times M}} \end{bmatrix}. \end{aligned}$$

If the individual entries k_{1i} are nonzero and the Jordan block diagonals have nonzero eigenvalues, the columns of Φ become linearly independent. Therefore, if $L \geq M$, the column rank of $\mathcal{O}_Y$ is M , which results in an observable system. To extend this proof to matrices $\hat{\mathbf{A}} = P\Lambda P^{-1}$, note that

$$\begin{aligned} \mathcal{O}_Y &= \begin{bmatrix} K\hat{\mathbf{A}}^{\tau_1} \\ \vdots \\ K\hat{\mathbf{A}}^{\tau_L} \end{bmatrix} \\ &= \begin{bmatrix} KP\Lambda^{\tau_1}P^{-1} \\ \vdots \\ KP\Lambda^{\tau_L}P^{-1} \end{bmatrix} \\ &= \hat{\mathbf{K}}P\Lambda^t P^{-1}, \end{aligned}$$

where $P \in \mathbb{R}^{ML \times ML}$, $\Lambda^t \in \mathbb{R}^{ML \times ML}$, and $P^{-1} \in \mathbb{R}^{ML \times ML}$ are the block diagonal matrices associated with the system. Since P is an invertible matrix, the conclusions about the column rank drawn before still hold, and the system is observable. ■

When the eigenvalues of the system matrix are repeated, it is not enough for K to be shaded. The next proposition takes a geometric approach and utilizes the rational canonical form of $\hat{\mathbf{A}}$ to obtain a lower bound on the number of required sampling locations. Let r be the number of unique eigenvalues of $\hat{\mathbf{A}}$, and let γ_{λ_i} denote the geometric

multiplicity of eigenvalue λ_i . Then, the *cyclic index* of $\hat{A}$ is defined as $\ell = \max_{1 \leq i \leq r} \gamma_{\lambda_i}$ [53].

Proposition 2

Suppose that the conditions in Proposition 1 hold, with the relaxation that the Jordan blocks $[\Lambda_1 \cdots \Lambda_O]$ may have repeated eigenvalues (that is, $\exists \Lambda_i$ and Λ_j , subject to $\lambda_i = \lambda_j$). Then, there exist kernels $k(x, y)$ such that the lower bound ℓ on the number of sampling locations N is given by the cyclic index of $\hat{A}$. In other words, the system in (3) is observable if $N \geq \ell$.

Proof by Contrapositive

It is shown that if the number of sampling locations is $N = \ell - 1$ (that is, $N < \ell$), then the system is not observable. Select the Gaussian kernel in the dictionary of atoms framework with sampling locations $x_i \in \mathcal{X}$ and that centers $c_j \in \mathcal{C}$, with the additional property that $x_i \neq x_j \forall i, j \in \{1, \dots, N\}$, $i \neq j$. In this case, $\mathbf{K}$ has $\ell - 1$ nonzero, linearly independent rows, and it can be written as

$$\mathbf{K} = \begin{bmatrix} k_{11} & k_{12} & \cdots & k_{1M} \\ \vdots & \vdots & \cdots & \vdots \\ k_{(\ell-1)1} & k_{(\ell-1)2} & \cdots & k_{(\ell-1)M} \end{bmatrix}.$$

The fact that the cyclic index is ℓ implies that at least one eigenvalue, λ , has ℓ Jordan blocks. Define indices $j_1, j_2, \dots, j_\ell \in \{1, 2, \dots, M\}$ as the columns corresponding to the leading entries of the ℓ Jordan blocks corresponding to λ . Without a loss of generality, let $j_1 = 1$. Using ideas similar to the previous proof, we can write the observability matrix as

$$\mathcal{O}_Y := \begin{bmatrix} k_{11}\lambda^{\tau_1} & \cdots & k_{1j_\ell}\lambda^{\tau_1} & \cdots \\ \vdots & \ddots & \vdots & \ddots \\ k_{11}\lambda^{\tau_\ell} & k_{1j_\ell}\lambda^{\tau_\ell} & \cdots & \cdots \\ \vdots & \vdots & \ddots & \ddots \\ k_{(\ell-1)1}\lambda^{\tau_1} & k_{(\ell-1)j_\ell}\lambda^{\tau_1} & \cdots & \cdots \\ \vdots & \vdots & \ddots & \ddots \\ k_{(\ell-1)1}\lambda^{\tau_\ell} & \cdots & k_{(\ell-1)j_\ell}\lambda^{\tau_\ell} & \cdots \end{bmatrix}.$$

Define $\boldsymbol{\lambda} := [\lambda^{\tau_1} \lambda^{\tau_2} \cdots \lambda^{\tau_\ell}]^T$. The preceding matrix becomes

$$\mathcal{O}_Y := \begin{bmatrix} k_{11}\boldsymbol{\lambda} & \cdots & k_{1j_2}\boldsymbol{\lambda} & \cdots & k_{1j_\ell}\boldsymbol{\lambda} & \cdots \\ \vdots & \ddots & \vdots & \ddots & \vdots & \ddots \\ k_{(\ell-1)1}\boldsymbol{\lambda} & \cdots & k_{(\ell-1)j_2}\boldsymbol{\lambda} & \cdots & k_{(\ell-1)j_\ell}\boldsymbol{\lambda} & \cdots \end{bmatrix}.$$

It must be shown that one of the columns can be written in terms of the others. This is equivalent to solving the linear system

$$\begin{bmatrix} k_{1j_1} \\ k_{2j_1} \\ \vdots \\ k_{(\ell-1)j_1} \end{bmatrix} = \begin{bmatrix} k_{1j_2} & \cdots & k_{1j_\ell} \\ k_{2j_2} & \cdots & k_{2j_\ell} \\ \vdots & \ddots & \vdots \\ k_{(\ell-1)j_2} & \cdots & k_{(\ell-1)j_\ell} \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_{\ell-1} \end{bmatrix}.$$

Since the kernel matrix on the right-hand side is generated from the Gaussian kernel from [54], it is known that every

principal minor of a Gaussian kernel matrix is invertible, which implies that $\mathcal{O}_Y$ cannot be observable. ■

A concrete example will be given to build intuition regarding this lower bound. For now, note the following proposition.

Proposition 3

Given the conditions stated in Proposition 2, it is possible to construct a measurement map $\tilde{K} \in \mathbb{R}^{\ell \times M}$ for the system given by (3) such that the pair $(\tilde{K}, \hat{A})$ is observable.

Proof

The construction of the measurement map $\tilde{K}$ is based on the rational canonical structure of $\hat{A}^T$, which decomposes $\mathcal{V}$ into $\hat{A}^T$ -cyclic direct summands such that $\mathcal{V} = \mathcal{V}_1 \oplus \dots \oplus \mathcal{V}_\ell$ (where ℓ is the cyclic index of $\hat{A}$). Let ξ_v be the MP of v (relative to $\hat{A}^T$); it is, then, the unique monic polynomial of the least degree such that $\xi_v(\hat{A}^T)v = 0$. Let $\alpha_1(\lambda)$ be the MP of $\hat{A}_{|\mathcal{V}_1}^T$; then, $\deg(\alpha_1(\lambda)) < M$. By the rational canonical structure theorem [53], there exists a vector $\tilde{v}_1$ such that $\xi_{\tilde{v}_1}(\lambda) = \alpha_1(\lambda)$. Similarly, there exists a vector $\tilde{v}_2$ such that $\xi_{\tilde{v}_2}(\lambda) = \alpha_2(\lambda)$ [where $\alpha_2(\lambda)$ is the MP of $\hat{A}_{|\mathcal{V}_2}^T$]. Thus, ℓ vectors are obtained that form the measurement map $\tilde{K} = [\tilde{v}_1, \tilde{v}_2, \dots, \tilde{v}_\ell]^T$. The construction of these vectors $\tilde{v}_i$ can be simplified by first performing the Jordan decomposition as $\hat{A}^T = P\Lambda P^{-1}$. Then, the vectors $\tilde{v}_i, i \in \{1, \dots, \ell\}$ for Λ can be constructed such that the entries corresponding to the leading entries of Jordan blocks of $\Lambda_{|\mathcal{V}_i}$ are nonzero. Such a construction ensures that the MP of vector $\tilde{v}_i$ with respect to $\Lambda_{|\mathcal{V}_i}$ is also the corresponding MP of $\Lambda_{|\mathcal{V}_i}$. Finally, the required map can be obtained as $\tilde{K} = [\tilde{v}_1, \tilde{v}_2, \dots, \tilde{v}_\ell]^T P^{-1}$. ■

The construction provided in the proof of Proposition 3 is utilized in Algorithm 1, which uses the rational canonical structure of $\hat{A}$ to generate a series of vectors $v_i \in \mathbb{R}^M$ whose iterations $\{v_1, \dots, \hat{A}^{m_1-1}v_1, \dots, v_\ell, \dots, \hat{A}^{m_\ell-1}v_\ell\}$ generate a basis for $\mathbb{R}^M$. Unfortunately, the measurement map $\tilde{K}$, being an abstract construction unrelated to the kernel, does not directly select $\mathcal{X}$. It is shown how to use the measurement map to guide a search for $\mathcal{X}$ in Remark 2. For now, a sufficient condition for observability of a general system is stated.

Theorem 1

Suppose that the conditions in Proposition 1 hold, with the relaxation that the Jordan blocks $[\Lambda_1 \cdots \Lambda_O]$ may have repeated eigenvalues. Let ℓ be the cyclic index of $\hat{A}$. Define

$$\mathbf{K} = [K^{(1)T} \cdots K^{(\ell)T}]^T \quad (8)$$

as the ℓ -shaded matrix [see Figure 4(b)], which consists of ℓ shaded matrices with the property that any subset of ℓ columns in the matrix is linearly independent from the others. Then, system (3) is observable if Y has distinct values and $|Y| \geq M$.

Proof

A cyclic index of ℓ for this system implies that there exists an eigenvalue λ that's repeated ℓ times. The theorem is proved for repeated eigenvalues of dimension 1. The same statement can be proved for repeated eigenvalues for Jordan blocks using the ideas in the proof of Proposition 1. Without a loss of generality, let $\mathbf{K}$ have ℓ fully shaded, linearly independent rows, and assume that the column indices corresponding to this eigenvalue are $\{1, 2, \dots, \ell\}$. Define $\lambda_i := [\lambda_i^{r_1} \ \lambda_i^{r_2} \ \dots \ \lambda_i^{r_\ell}]^T$. Then,

$$\mathcal{O}_Y := \begin{bmatrix} k_{11}\lambda_1 & k_{12}\lambda_2 & \dots & k_{1M}\lambda_M \\ \vdots & \vdots & \ddots & \vdots \\ k_{\ell 1}\lambda_1 & k_{\ell 2}\lambda_2 & \dots & k_{\ell M}\lambda_M \end{bmatrix}.$$

Let $\lambda_1 = \lambda_2 = \dots = \lambda_\ell = \lambda$. Focusing on these first ℓ columns of this matrix, this implies that constants $c_1, c_2, \dots, c_{\ell-1}$ must be determined such that

$$\begin{bmatrix} k_{11} \\ \vdots \\ k_{\ell 1} \end{bmatrix} = c_1 \begin{bmatrix} k_{12} \\ \vdots \\ k_{\ell 2} \end{bmatrix} + \dots + c_{\ell-1} \begin{bmatrix} k_{1\ell} \\ \vdots \\ k_{\ell \ell} \end{bmatrix}.$$

However, these columns are linearly independent by assumption, and thus no such constants exist (implying that $\mathcal{O}_Y$ is observable). ■

While Theorem 1 is quite a general result, the condition that any ℓ columns of $\mathbf{K}$ be linearly independent is a very stringent condition. One scenario where this condition can be met with minimal measurements is when the feature map $\hat{\psi}(x)$ is generated by a dictionary of atoms with the Gaussian RBF kernel evaluated at sampling locations $\{x_1, \dots, x_N\}$ according to (2) (where $x_i \in \Omega \subset \mathbb{R}^d$), while x_i are sampled from a nondegenerate probability distribution on Ω , such as the uniform distribution. For a semideterministic approach, when the dynamics matrix $\hat{A}$ is block diagonal, the following simple heuristic may be employed.

Remark 2

Let Ω be compact, $C = \{c_1, \dots, c_M\}$, $c_i \in \Omega$, and the approximate feature map be defined by (2). Consider the system

ALGORITHM 2 Sampling locations set $\mathcal{X}$.

Input: $\hat{A} = C$, lower bound ℓ
 Decompose C to generate invariant subspaces $\hat{\mathcal{H}}_j, j \in \{1, 2, \dots, \ell\}$ (see the “Preliminaries of Rational Canonical Structures” section)
for $j = 1$ to ℓ **do**
 Obtain centers $C^{(j)}$ with respect to subspace $\hat{\mathcal{H}}_j$,
 Generate samples $x_i^{(j)}$ to create a kernel matrix $K^{(j)}$ that is shaded only with respect to centers $C^{(j)}$
end for
Output: Sampling locations set $\mathcal{X} = \{x^{(1)}, x^{(2)}, \dots, x^{(\ell)}\}$.

(3) with $\hat{A} = \Lambda$, and let $Y = \{0, 1, \dots, M-1\}$. Then, the measurement map $\tilde{K}$'s values lie in $\{0, 1\}$; specifically, each row $\tilde{K}^{(j)}, j \in \{1, \dots, \ell\}$ corresponds to a subspace $\hat{\mathcal{H}}_j$ generated by a subset of centers $C^{(j)} \subset C$. Generate samples $x_i^{(j)}$ to create a kernel matrix $K^{(j)}$ that is shaded only with respect to centers $C^{(j)}$. Once this is completed, move on to the next subspace $\hat{\mathcal{H}}_{j+1}$. When all ℓ rows of $\tilde{K}$ are accounted for, construct the matrix $\mathbf{K}$, as in (8). Then, the resulting system $(\mathbf{K}, \hat{A})$ is observable.

This heuristic is formalized in Algorithm 2. Note that, in practice, the matrix $\hat{A}$ must be inferred from measurements of the process f_τ . If no assumptions are placed on $\hat{A}$, it is clear that at least M sensors are required for the system identification phase. Future work will study the precise conditions under which system identification is possible with fewer than M sensors.

DISCUSSION OF THEORETICAL RESULTS

The systems-theoretic approach in this article reveals something rather surprising: functions with complex dynamics (with a small cyclic index) can be recovered with fewer sensor placements than functions that have simpler dynamics. Although seemingly counterintuitive, it becomes clear that this is because complex dynamics, which are characterized by a lower geometric multiplicity of the eigenvalues, ensure that the orbit $\Theta := \{\lambda w_\tau\}_{\tau \in Y}$ traverses a greater portion of $\mathbb{R}^M \equiv \hat{\mathcal{H}}$, and thus fewer sensors can recover more geometric information. In simpler functional evolution, Θ develops along strict subspaces of $\mathbb{R}^M$. Thus, more independent sensors are required to infer the same amount of information.

In the case described in Remark 2, we have a set of centers $C = \{c_1, \dots, c_M\}$, which generates the bases $\mathcal{F}^C = \{\psi(c_1), \dots, \psi(c_M)\}$. Let the cyclic index be (ℓ) . This implies that there exist ℓ subsets $\Psi^{(j)}$ of $\mathcal{F}^C$ with at least one element $\psi(c_j)$ each, leading to $\binom{M}{\ell}$ possible choices. Figure 8 represents these choices as hyperplanes separating the subsets. The measurement map described in [21, Algorithm 1] induces this *decomposition of bases* $\mathcal{F}^C = \{\Psi^{(1)}, \dots, \Psi^{(\ell)}\}$ in polynomial time. Furthermore, each subset $\Psi^{(j)}$ is directly associated to a subset of centers $C^{(j)} \subset C$, which enables selecting targeted sensor locations $x_i \in \Omega$. Specifically, for radially symmetric kernels, such as Gaussian, the centroid of the convex hull of $C^{(j)}$ is sufficient for generating a sensor placement. The measurement map is a significant theoretical insight into sensor placement for dynamically changing environments because it directly takes into account the dynamics of the process. Of course, in practice, this may be too expensive for approximate feature spaces where M is very large. Thus, random sampling can be used to generate the sensor locations, instead, at the cost of N being larger than ℓ . The advantage is that, since random sampling is computationally inexpensive, different choices of sensor placements can be generated and evaluated relatively quickly.

Another point to note is that since the collection of bases $\{\hat{\psi}_i(x)\}_{i=1}^M$ determines the richness of the function space $\hat{\mathcal{H}} \approx \mathcal{H}$ we operate in, it governs the fidelity of the model approximation to the true time-varying function. As a consequence, observability of the system in $\hat{\mathcal{H}}$ refers to the best possible approximation in $\hat{\mathcal{H}}$. The greater the number of bases, the higher the dimensionality (which results in greater model fidelity but may require a much greater number of measurements for state recovery). This is where the lower bounds presented in the article are particularly useful. They show that for functional evolutions corresponding to a certain $\hat{A}$, the number of sensor placements is essentially independent of the dimensionality M but depends, rather, on the cyclic index of $\hat{A}$. Figure 7 gives an overall picture of the process of generating a kernel observer, while Figure 8 displays two approaches to sensor selection in our framework. The measurement map approach can generate a smaller set of sensors than the random placement approach but comes at an additional computational cost.

RANDOM SENSOR PLACEMENT

We now elaborate on how the challenging problem of sensor placement can be addressed through random selection. The process of random selection is a product of the kernel observer model described previously. The theoretical background required to prove Theorem 2 (which states the expected number of randomly placed sensors required to monitor a given spatiotemporal process) and Theorem 3 (which determines the probability with which optimal sensor placement is ensured) is presented, given that N number of sensors has been placed. As discussed earlier, an approximate feature space $\hat{\mathcal{H}}$ is used, with the corresponding transition operator $\hat{A}: \hat{\mathcal{H}} \rightarrow \hat{\mathcal{H}}$ representing finite-dimensional functional evolution.

To achieve observability for the pair $(\hat{A}, K)$, row vectors of the corresponding observability matrix $\mathcal{O}$ should form the basis for the $\mathbb{R}^M$ -dimensional space $\hat{\mathcal{H}}$. According to the rational canonical structure theorem [53], $\hat{A}$ can successively decompose the dual space $\mathbb{R}^M$ into subspaces, $\mathcal{V}_i \subset \mathcal{V}, i \in \{1, \dots, \ell\}$, which have the following properties: 1) $\mathcal{V} = \mathcal{V}_1 \oplus \dots \oplus \mathcal{V}_\ell$, 2) $\hat{A}\mathcal{V}_i \subset \mathcal{V}_i$, and 3) $\hat{A}|_{\mathcal{V}_i}, i \in \{1, \dots, \ell\}$ are cyclic. The integer ℓ is unique and called the *cyclic index* of $\hat{A}$. Each of these properties contributes to the theorem related to the number of random samples required to achieve observability. The first property shows that the space $\mathbb{R}^M$ can be decomposed into ℓ independent subspaces. The second property shows that the vector $v_i \in \mathcal{V}_i$ stays in $\mathcal{V}_i$, even when operated on by $\hat{A}$. To generate bases for $\mathbb{R}^M$, at least ℓ vectors $v_1, \dots, v_\ell$ are needed, where each v_i arises from a corresponding subspace $\mathcal{V}_i$ from the set of subspaces $\mathcal{V}_1, \dots, \mathcal{V}_\ell$. This holds due to the third property, but it requires that the vectors $v_1, \dots, v_\ell$ are the cyclic generators of their corresponding subspaces.

This analysis is based on whether a randomly selected sensor can generate a cyclic generator. To examine this, recall that a row vector $K_{(i)}$ generated by a randomly selected sensor location x_i takes the form

$$K_{(i)} = [k(x_i, c_1), \dots, k(x_i, c_M)]. \quad (9)$$

For radial kernels, for example, the entries corresponding to the centers closer to x_i tend to be nonzero, whereas the others tend to be zero.

Overall, our construction is as follows. For each subspace $\mathcal{V}_i$, let $C_{\mathcal{V}_i} \subset C$ be the centers corresponding to the leading entries of the Jordan blocks. The minimum number of random samples required to generate the bases for $\mathcal{V}_i$ is equal to the number of Jordan blocks comprising $\mathcal{V}_i$. Altogether, the minimum number of random samples required to generate a basis for $\mathbb{R}^M$ is equal to the total number of Jordan blocks in $\hat{A}$. Let s be the total number of Jordan blocks in $\hat{A}$. Then,

$$s = \sum_{\lambda \in \sigma(\hat{A})} \gamma_{\hat{A}}(\lambda), \quad (10)$$

where $\sigma(\hat{A})$ represents the spectrum of $\hat{A}$ whose elements are the eigenvalues of $\hat{A}$ and $\gamma_{\hat{A}}(\lambda)$ is the geometric multiplicity corresponding to the eigenvalue λ (which is also

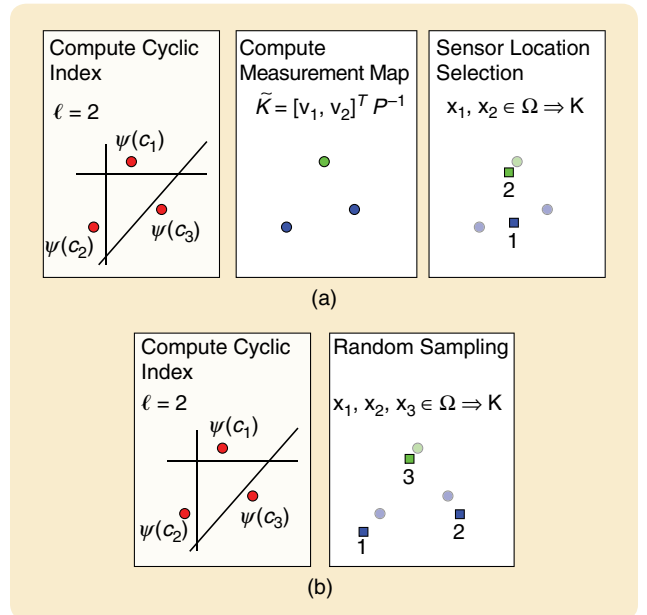


FIGURE 8 Sensor placement using the measurement map and random sampling approaches. The circles represent data locations associated to bases [such as $c_i \leftrightarrow \psi(c_i)$], and the squares indicate sensor locations [such as $x_i \leftrightarrow \psi(x_i)$]. The cyclic index ($\ell = 2$) shows how many possible couplings of bases exist, which can be represented as a choice of $\binom{M}{\ell}$ hyperplanes in Ω . (a) If the measurement map is computed, the correct couplings are chosen (green versus blue), and a smaller number of sensors (two) can be placed. (b) Alternatively, random sampling is more computationally efficient but generally requires more sensors (three).

Comparing this to the spectral expansion of $\hat{u}(f, x)$ (and letting $\lambda_1 = 1$),

- 3) $\hat{u}^*(f, x) = \hat{\Psi}(x) \cdot v_1$, $\hat{s}_j(x) = \hat{\Psi}(x) \cdot v_j$, and there is no continuous spectrum component. The hypothesis now follows from Proposition 5. ■

Ultimately, a direct connection between the spectral decomposition of the transition matrix in the dual space of the approximate feature space and the Koopman modes, eigenfunctions, and eigenvalues was established.

Corollary 1

Suppose that given $\epsilon > 0$, an approximate feature space $\hat{\mathcal{H}}$ was determined such that for all $f \in \mathcal{H}$, there exists $\hat{f} \in \hat{\mathcal{H}}$ such that $\|f - \hat{f}\| < \epsilon$. If $\phi_j(f) \neq 0$, there exists an approximate feature space such that $\hat{\Psi}(x) \cdot v_j$ is arbitrarily close to the Koopman mode $s_j(x)$ (where v_j is an eigenvector of the dual space transition matrix $\hat{A}$).

Proof

Since ϕ_j is continuous, then $|\phi_j(f) - \phi_j(\hat{f})|$ can be made arbitrarily small. This means that if $\phi_j(f) \neq 0$, then $(\phi_j(f)/\phi_j(\hat{f}))$ can be made arbitrarily close to one. From Theorem 4 and the fact that $s_j \in \mathcal{H}$ means s_j is bounded, it is concluded that $\hat{\Psi}(x) \cdot v_j$ can be made arbitrarily close to $s_j(x)$ everywhere in Ω . ■

This implies that if a close-enough approximate feature space can be determined, then the eigenvectors of the transition matrix in the dual space correspond exactly with the Koopman modes in the input space.

INVARIANT SUBSPACES

One way of viewing the invariant subspaces concept is to say that information contained in an invariant subspace never leaves that invariant subspace. It is hypothesized that the kernel centers associated with the invariant subspaces of a spatiotemporally evolving system are generally associated with spatial regions in the domain (and not just homogeneously spread all over and mixed with the other invariant subspaces). This hypothesis makes sense both physically and mathematically. In physics, the *principle of locality* states that an object is only directly influenced by its immediate surroundings [59]. If this is the case (and there is no reason to believe that it is not, apart from certain quantum dynamics situations) and the E-GP model accurately captures the physics of the system, then information (measurable phenomena) may continuously travel only from one point in the domain to another. Mathematically, since a value at any one point in the domain is influenced by the weights of multiple nearby centers, proximate centers are expected to be dynamically connected unless separated by “plains” (where the values are indistinguishable from noise). In an ideal case, the Jordan form of an $n \times n$ matrix $\hat{A}$ is block diagonal and therefore gives a decomposition of the n -dimensional Euclidean space into clearly separated invariant subspaces of $\hat{A}$. The cyclic index, which can be

found by counting the geometric multiplicities of eigenvalues in $\hat{A}$, gives the number of invariant subspaces.

In practice, data-driven approximations of $\hat{A}$ rarely give such easily interpretable properties. This fact drives the need for an algorithm that can divide the system into invariant subspaces, even when the boundaries between such spaces are not mathematically precise. This algorithm can be thought of as a soft Jordan decomposition. Each block in the normal Jordan form has a set of corresponding eigenvectors and an eigenvalue with geometric multiplicity. When the former is transformed back into the domain space, complex-valued functions are obtained that are the Koopman modes of the system. These provide an image of what kind of structures are seen in the dynamics. The eigenvalues describe the frequency with which these structures oscillate between their real and imaginary forms as well as their exponential growth or decay in magnitude.

k-INVARIANT SUBSPACE CLUSTERING

In response to the need for an algorithm that can separate invariant subspaces in the linear model of a system generated by data, *k-invariant subspaces clustering* was developed. The intuition behind this algorithm is to replace the Euclidean distance in the well-known *k*-means algorithm with a different metric of “nearness,” namely one corresponding with the dynamic connections in the space. The $\hat{A}$ matrix provides easy access to these. Its rows $\hat{A}_{i*}$ indicate which centers inform the i th value of $w_{\tau+1}$, and its columns $\hat{A}_{*j}$ indicate which centers will be informed by the j th value of w_τ . However, these values tend to be biased toward the eigenmodes with higher frequencies (that is, those whose eigenvalues have a greater polar angle) and those that grow exponentially. To control for this, the eigenvalues of the matrix were modified as follows: 1) zero any eigenvalue that is clearly inside (not on) the unit circle, 2) unitize the remaining eigenvalues, and 3) adjust the frequency of the remaining eigenvalues on the unit circle to either $\pm(\pi/4)$. If the eigendecomposition of the original was $\hat{A} = UDU^{-1}$, then construct a new matrix with the modified eigenvalues $\bar{D}$ as $\bar{A} = U\bar{D}U^{-1}$.

Much like the pairwise-squared deviations of the points formulation of the *k*-means clustering problem, the problem can be written as

$$\arg\max_S \sum_{i=1}^k \frac{1}{2|S_i|} \sum_{\substack{x_i \neq x_j \\ x_j, x_i \in S_i}} \bar{A}_{ij}^2.$$

This problem can be solved with Algorithm 3. Note the differences between this and the *k*-means clustering algorithm. First, we are *maximizing* since the terms represent influence rather than distance. Second, in the possible case that $|S_k| = 0$, the total value is set to zero to avoid division by zero. Third, exclude the centers’ influence on themselves

from the score by taking $S_k \setminus \{i\}$. This algorithm can be repeated as many times as desired, with different random initial conditions, and the result that produces the highest score is retained.

EMPIRICAL RESULTS

We begin with numerical results obtained for determining sensing locations through the application of kernel observers to the prediction of spatiotemporal functions. The highlight here is a set of results related to sensing global ocean

temperatures by using information from just a few locations, employing *Advanced Very-High-Resolution Radiometer (AVHRR)* satellite data. This is followed by an application of the E-GP in predicting spatiotemporal functions. Results were included that predicted weed growth in simulated agricultural fields. The application of the E-GP to model the flow past a cylinder at different Reynolds numbers is explored in “Learning Fluid Flows With Evolving Gaussian Processes.” The section concludes with a comparison of eigenvalues and Koopman modes computed from the E-GP to those of DMD.

ALGORITHM 3 The k -invariant subspaces algorithm.

```

while clusters have changed do
  for each center  $i$  do
    Find cluster  $k$  that maximizes the score
     $\frac{1}{|S_k|} (\|\bar{A}_{i, S_k \setminus \{i\}}\|^2 + \|\bar{A}_{S_k \setminus \{i\}, i}\|^2)$ 
    Reassign center  $i$  to cluster with highest score
  end for
end while
return clusters

```

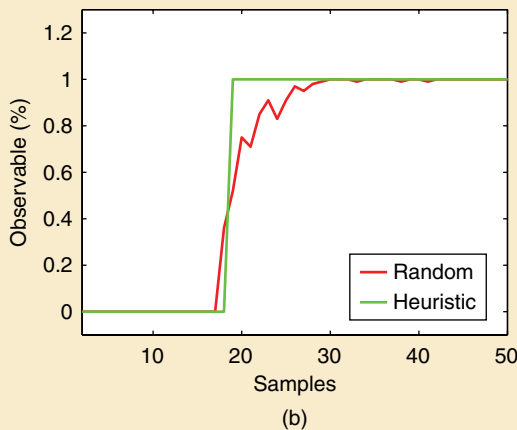
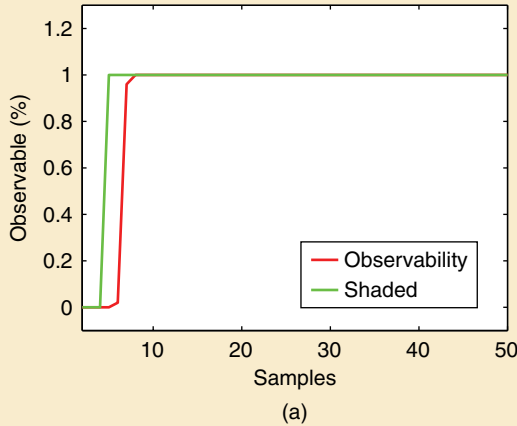


FIGURE 9 Kernel observability results. (a) Shaded versus observability. (b) Heuristic versus random.

Sensing Locations for Synthetic Data Sets

The goal of this experiment is to investigate the dependency of the observability of system (3) on the shaded observation matrix and the lower bound presented in Proposition 2. The domain is fixed on the interval $\Omega = [0, 2\pi]$. First, select sets of points $C^{(i)} = \{c_1, \dots, c_M\}$, $c_j \in \Omega$, and $M = 50$, and construct a dynamics matrix $A = \Lambda \in \mathbb{R}^{M \times M}$ with cyclic index 5. Select the RBF kernel $k(x, y) = e^{-\|x - y\|^2 / 2\sigma^2}$, $\sigma = 0.02$. Generating samples $\mathcal{X} = \{x_1, \dots, x_N\}$, $x_i \in \Omega$ randomly, compute the ℓ -shaded property and observability for this system. Figure 9(a) shows how shadedness is a necessary condition for observability, validating Proposition 1: the slight gap between shadedness and observability can be explained due to numerical issues in computing the rank of O_T .

Next, consider a system with a cyclic index $\ell = 18$ to verify random-sensor placement results. The measurement operator K was constructed using the heuristic in Remark 2 (Algorithm 2) and random-sensor selection to generate the sampling locations $\mathcal{X}$. These results are presented in Figure 9(b). The plot for random sampling, which has been averaged across 100 runs, resembles a cumulative distribution function of an exponential distribution $F(X = x) = 1 - \exp(-\lambda x)$. This verifies the claim made in Theorem 3, as the probability of becoming unobservable decays exponentially with the number of placed sensors. Also, fitting an exponential distribution, it was determined that the mean λ^{-1} comes close to the ratio ς/p , which is the expected number of randomly placed sensors required for observability, as per Theorem 2. Note that observability is not achieved if the number of samples $N < \ell$, which is experimental evidence of the result in Proposition 2.

Comparison With Nonstationary Kernel Methods on Real-World Data

Three real-world data sets were used to evaluate and compare the kernel observer with the two different lines of approach for nonstationary kernels discussed in the related work. The process convolution with a local smoothing kernel (PCLSK) and the latent extension of the input space (LEIS) are compared with the nonstationary space-time variable latent length GP in [8] and [37], respectively, using Intel Berkeley, Irish wind, and ozone data sets. The model

inference for the kernel observer involved the following three steps:

- 1) Selecting the Gaussian RBF kernel $k(x, y) = e^{-\|x-y\|^2/2\sigma^2}$, a search for the ideal σ is performed for a sparse GP model (with a fixed basis vector set C selected using the method in [56]). For the data set discussed in this section, the number of basis vectors was equal to that of the sensing locations in the training set, with the domain for the input set defined across $\mathbb{R}^2$.
- 2) Having obtained σ , GP inference is used to generate weight vectors for each time step in the training set, resulting in the sequence $w_\tau, \tau \in \{1, \dots, T\}$.
- 3) Matrix least squares is applied to this sequence to infer $\hat{A}$ (Algorithm 4).

ALGORITHM 4 The kernel observer (transition learning).

Input: Kernel k , basis centers C , final time step T .

while $\tau \leq T$ **do**

- 1) Sample data $\{y_\tau^i\}_{i=1}^M$ from f_τ .
- 2) Solve for $\hat{w}_\tau$ by using least squares on $y_\tau = K\hat{w}_\tau$.
- 3) Store weights $\hat{w}_\tau$ in matrix $\mathcal{W} \in \mathbb{R}^{M \times T}$.

end while

To infer $\hat{A}$, define matrix $\Phi = \mathcal{W}^T \mathcal{W}$. Then:

for $i = 1$ to M **do**

At step i , solve system

$$\hat{A}^{(i)} = ((\Phi + \lambda I)^{-1} (\mathcal{W}^T \mathcal{W}^{(i)}))^T, \quad (17)$$

where $\hat{A}^{(i)}$ and $\mathcal{W}^{(i)}$ are the i th columns of $\hat{A}$ and $\mathcal{W}^{(i)}$, respectively.

end for

Compute the covariance matrix $\hat{B}$ of the observed weights $\mathcal{W}$.

Output: estimated transition matrix $\hat{A}$, predictive covariance matrix $\hat{B}$.

ALGORITHM 5 The kernel observer (monitoring and prediction).

Input: Kernel k , basis centers C , estimated system matrix $\hat{A}$, estimated covariance matrix $\hat{B}$.

Compute Observation Matrix: Compute the cyclic index ℓ of $\hat{A}$, and compute K .

Initialize Observer: Use $\hat{A}, \hat{B}$, and K to initialize a state observer (such as a Kalman filter) on $\hat{\mathcal{F}}_\ell$.

while measurements available **do**

- 1) Sample data $\{y_\tau^i\}_{i=1}^N$ from f_τ .
- 2) Propagate Kalman filter estimate $\hat{w}_\tau$ forward to time $\tau + 1$, correct using measurement feedback with $\{y_{\tau+1}^i\}_{i=1}^N$.
- 3) Output predicted function $\hat{f}_{\tau+1}$ of Kalman filter.

end while

For prediction in the autonomous setup, $\hat{A}$ is used to propagate the state w_τ forward to make estimates with no feedback, and in the observer setup, a Kalman filter (Algorithm 5), with N determined using Proposition 2 (and locations picked randomly), is used to propagate w_τ forward to make forecasts. This is also compared with a baseline GP (denoted by *original GP*), which is the sparse GP model trained using all the available data.

The first data set, the Intel Berkeley research lab temperature information, consists of 50 wireless temperature sensors in an indoor laboratory region spanning 40.5 m in length and 31 m in width (<http://db.csail.mit.edu/labdata/labdata.html>). The training data consists of temperature information from March 6, 2004 at intervals of 20 min (beginning 12:20 a.m.), which totals to 72 time steps. Testing is performed across another 72 time steps, beginning 12:20 p.m. on the same day. Out of 50 locations, 25 were uniformly selected for training and testing purposes. Results of the prediction error are shown as a box plot in Figure 10(a) and as a time series in Figure 10(b). Note that, in the figure,

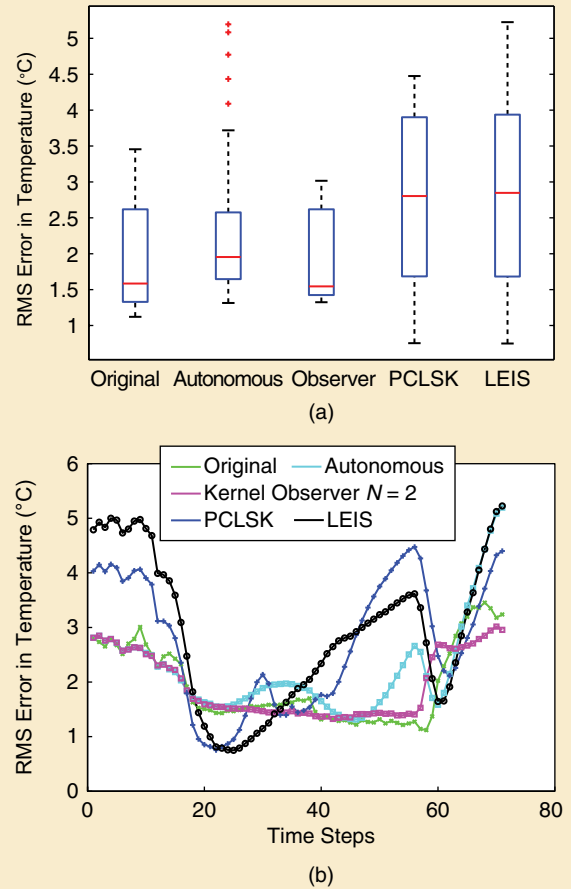


FIGURE 10 A comparison of the kernel observer versus the process convolution with a local smoothing kernel (PCLSK) and the latent extension of the input space (LEIS) on the Intel Berkeley data set. The error as (a) a box plot and (b) a time series. RMS: root mean square.

Auto refers to autonomous setup. The cyclic index of $\hat{A}$ was determined to be two, so N was set to two for the kernel observer with feedback. Note that even the autonomous kernel observer outperforms the PCLSK and the LEIS, overall, and the kernel observer with feedback with $N = 2$ significantly outperforms all the other methods (which is why results with $N > 2$ were not included).

The second data set contains wind information consisting of daily average wind speed measurements collected, from 1961 to 1978, at 12 meteorological stations in the Republic of Ireland (<http://lib.stat.cmu.edu/datasets/wind.desc>). The prediction error is given as a box plot in Figure 11(a) and as a time series in Figure 11(b). Again, the cyclic index of $\hat{A}$ was determined to be two. In this case, the autonomous kernel observer's performance is comparable to the PCLSK and LEIS, while the kernel observer with feedback with $N = 2$ again outperforms all the other methods.

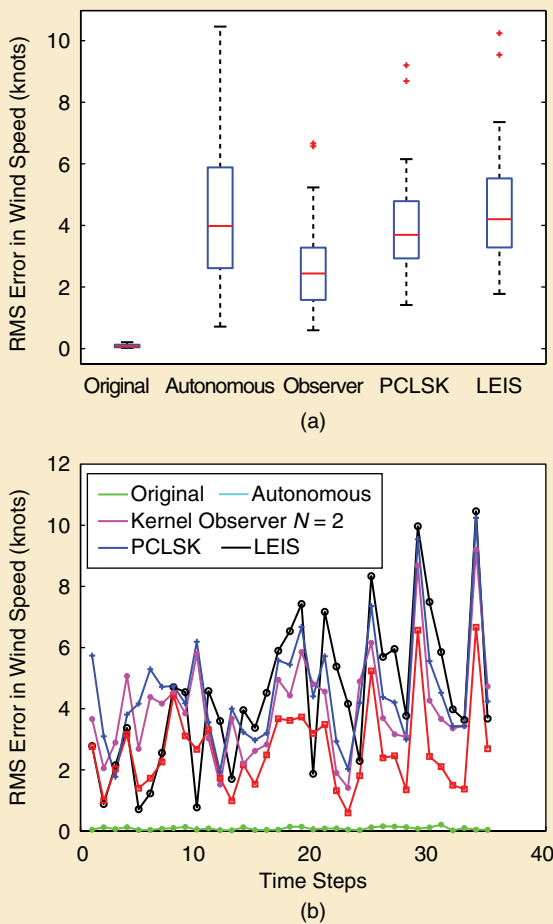


FIGURE 11 A comparison of the kernel observer versus the process convolution with a local smoothing kernel and the latent extension of the input space on the Irish wind data set. The error as (a) a box plot and (b) a time series. RMS: root mean square.

Finally, the third data set records the ozone concentration (in parts per billion) measured at 60 stations across the country by the United States Environmental Protection Agency [60]. Due to missing measurements, only data from 1997 to 2013 are selected for training and evaluation. Each station averaged the ozone concentration during a period of three months, resulting in four quarters per year. Out of 60 sensor locations, 30 were uniformly selected for training, and the remaining stations were chosen for testing purposes. The prediction error results are presented as a box plot in Figure 12(a) and as a time series in Figure 12(b). Here, the cyclic index of $\hat{A}$ was determined to be one. In this case, the performance of the autonomous kernel observer is comparable to the PCLSK and LEIS, with the kernel observer

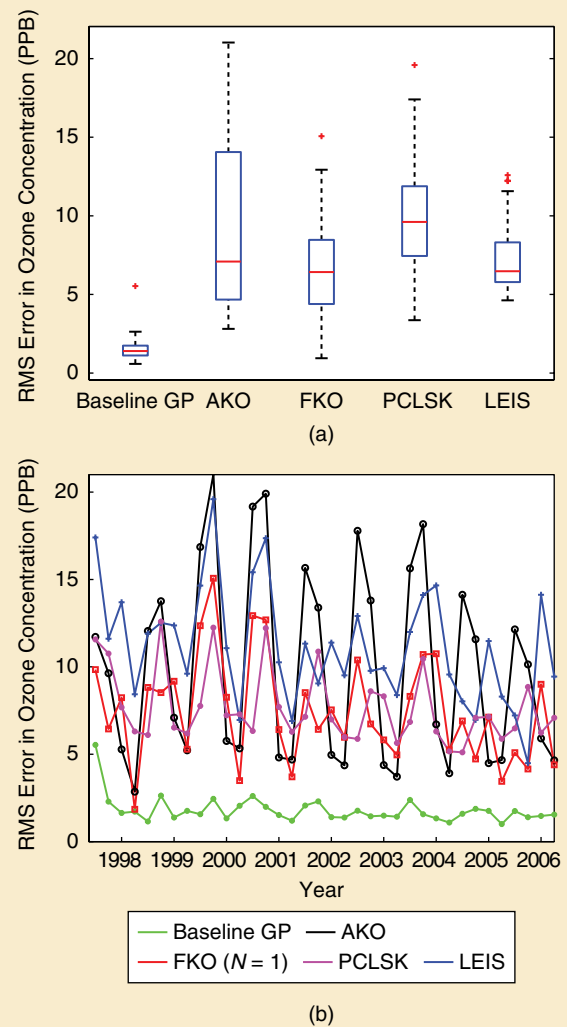


FIGURE 12 A comparison of the kernel observer versus the process convolution with a local smoothing kernel and the latent extension of the input space using the ozone data set. The error as (a) a box plot and (b) a time series. RMS: root mean square; PPB: parts per billion; GP: Gaussian process; AKO: autonomous kernel observer; FKO: feedback kernel observer; PCLSK: process convolution with a local smoothing kernel; LEIS: latent extension of the input space.

with feedback with $N = 1$ performing the best. Table 1 reports the total training and prediction times associated with the PCLSK, LEIS, and kernel observer. It was observed that 1) the kernel observer is an order of magnitude faster than the competing methods and 2) even for small sets, competing methods did not scale well.

Prediction of Global Ocean Surface Temperature

The feasibility of this approach was analyzed on a very large data set from the National Oceanographic Data Center: the 4-km AVHRR Pathfinder Project, which monitors the global ocean surface temperature [Figure 13(a) provides the raw data]. This data set is challenging, with measurements at more than 37 million possible coordinates but with only 3–4 million measurements available per day (which means a lot of missing data). The goal was to learn the day and night temperature models by using data from 2011 and then monitor the system through 2012. Success in monitoring would demonstrate two things: 1) the modeling process can capture spatiotemporal trends that generalize across years and 2) the observer framework facilitates inferring the state using a number of measurements that is an order of magnitude fewer than the available total. Note that the data set and the nonstationary kernel methods were not compared due to their size and high computational requirement.

To build the autonomous kernel observer and general kernel observer models, the same procedure outlined in the “Comparison With Nonstationary Kernel Methods on Real-World Data” section was followed but with $C = \{c_1, \dots, c_M\}$, $c_j \in \mathbb{R}^2$, and $|C| = 300$. The Kalman filter for the general kernel observer model used $N \in \{250, 500, 1000\}$ at random locations to track the system state, given a random initial condition w_0 . As a fair baseline, the observers are compared to training a sparse GP model (labeled *original*) on approximately 400,000 measurements per day. Figure 13(b) estimates the global ocean surface temperatures obtained using the autonomous kernel observer.

In Figure 13, (a) and (d) compare the autonomous and feedback approach with 1000 samples to the baseline GP. It is shown that the autonomous method does well in the beginning but then incurs an unacceptable amount of error when the time series goes into 2012 (that is, where the model has not seen any training data), whereas the feedback kernel observer does well throughout. In Figure 13, parts (e) and (f) compare the root mean square error of estimated values from the real data. This figure shows the trend of the observer getting progressively better state estimates as a function of the number of sensing locations N . Note that the performance of training a GP was checked with only 1000 samples as a control. However, the average error was approximately 10 K (that is, much worse than the feedback kernel observer). The time required for using the kernel observer is significantly less than retraining the model at every time step [see Figure 13(g) and (h)].

Weather Anomaly in 2012

The poor performance of the autonomous kernel observer in 2012 was further investigated, as illustrated in Figure 13(c) and (d). Clearly, the prediction error blows up at the start of May 2012, indicating that the autonomous model trained using data from 2011 does well in capturing the annual weather dynamics up to that month. Attention is directed to the weather in May 2012, as changes in ocean temperatures are directly related to the weather. As assumed, severe weather conditions were reported on the east coast of the United States, and this anomaly continued until June. The apparent poor performance of the autonomous kernel observer can be a useful indicator in detecting anomalous behavior, as was the case with the ocean temperature in May 2012 (which deviated from the normal weather dynamics observed in 2011, when no severe conditions were reported).

The locations at which the prediction error was two standard deviations above the mean error were further identified. They are plotted in Figure 14. The error locations from May 6 to 7 coincided with the severe weather onset on the east coast, while the locations on May 28 and 29 were along the track of Tropical Storm Beryl. This shows that the autonomous kernel observer model captures the dynamics of spatiotemporal evolution and has the potential to identify current behavior that is anomalous to what is observed in the training set.

Predicting the Evolution of Weed Density in Agricultural Fields

As a proof of concept for applying our methods for learning dynamics and/or inferring the state of spatiotemporally evolving systems, the problem of predicting weed growth in agricultural fields was examined. Past work presented a detailed analysis of weed growth models in which measurements of the seed bank density for various species were conducted [61]–[64]. To generate data for testing our methods, a weed growth simulation model was implemented whose rate of seedling emergence agreed with that found in the research [1]. A Poisson process was utilized to simulate the temporal evolution of emergence events. This assumption is reasonable across

TABLE 1 Total training and prediction times for Figures 10 and 11.

	Intel Berkeley	Irish Wind	Ozone
Data size (bases/time steps)	25/72	12/36	30/68
Kernel observer	2.1 s	0.1 s	1.61 s
Process convolution with a local smoothing kernel	121.4 s	7 s	91.9 s
Latent extension of the input space	43.8 s	2.8 s	37.41 s

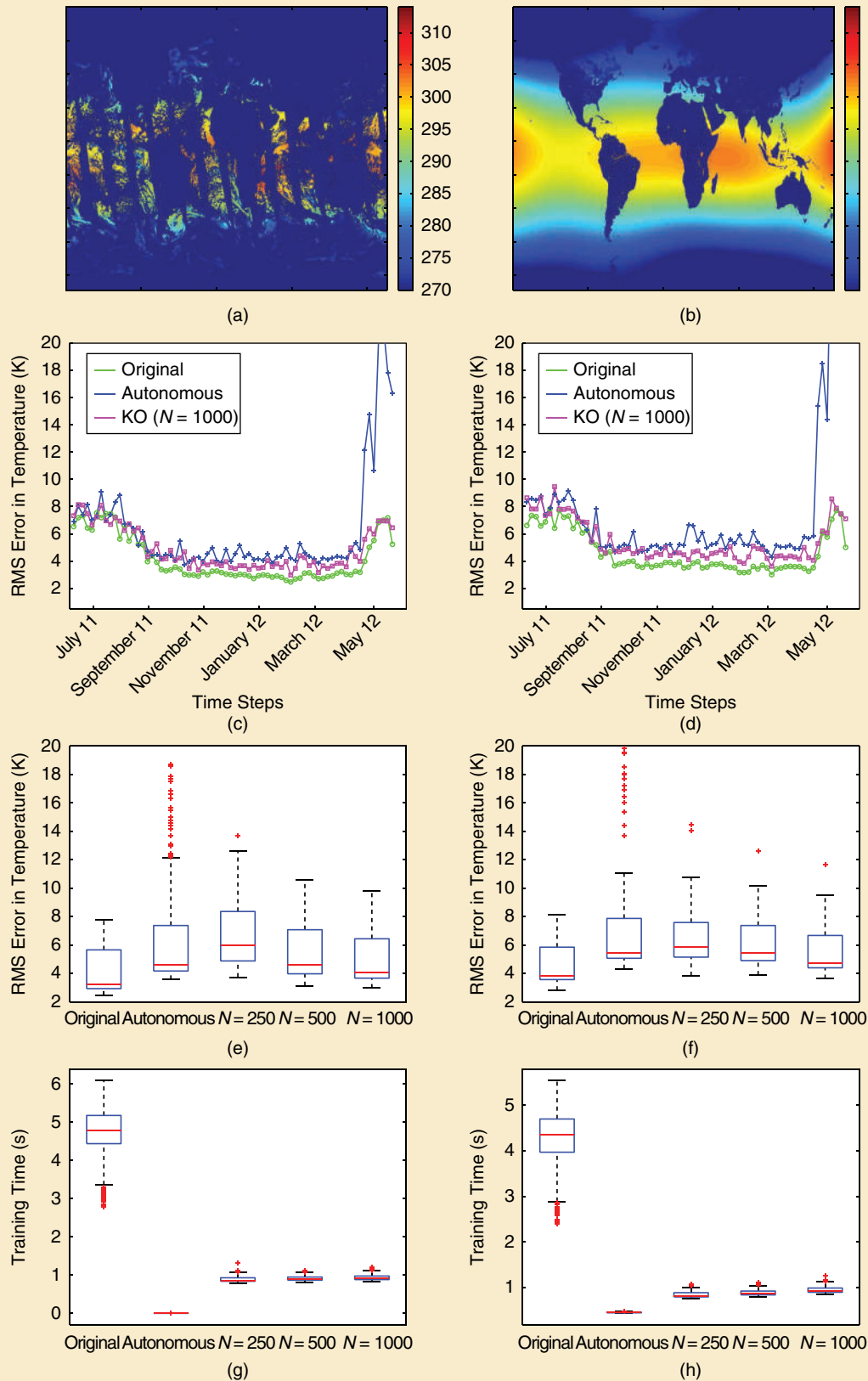


FIGURE 13 The performance of the kernel observer using *Advanced Very-High-Resolution Radiometer* satellite 2012 data with different numbers of observation locations. (a) The raw satellite data. (b) The autonomous kernel observer (KO) estimate. (c) The day error (time series). (d) The night error (time series). (e) The day error (box plot). (f) The night error (box plot). (g) The day estimation time. (h) The night estimation time. RMS: root mean square.

the short time scales during which robots may fully weed a field.

Other work in the field of crop science [65], [66] has shown that the spatial variation in the seed bank density for some species of weeds may be modeled via the Gini coefficient of concentration (GCC). This model is accurate for common waterhemp (*Amaranthus tuberculatus*), and thus it was found to be useful in this simulation of the spatial distribution of the seed bank density. See [1], [67], and [68] regarding the relationship between seed bank emergence patterns and environmental conditions such as temperature and moisture.

The weed growth simulation (Figure 15) is based on Bernoulli random variables, operating on a matrix of cells (each 0.8 m^2) comprising a gridded field of 0.4 hectares, or a cellular automata model [69].

Seeds emerge from a limited bank, forming a binomial distribution through time. The parameters that are used, which are summarized in Table 2, are aligned with the growth model for common waterhemp determined in [65]. The initial density of the seed bank in each cell is S_0 , on average (between 600 and 1560 seeds per cell). However, at the start of the simulation, the seed bank density in each cell $S_0(x, y)$ is chosen so that the GCC between all the cells (used to ensure that the relative density of the weeds aligns with that seen in real experiments) is from 0.31 to 0.35, as experimentally determined in [65]. The field is first divided into 50 patches of weeds, with centers chosen uniformly at random and with sizes from zero to 20 cells in each dimension. Each cell has an initial density between zero and 20. At the initialization of the simulation, a certain number of days (d_0) is allowed to

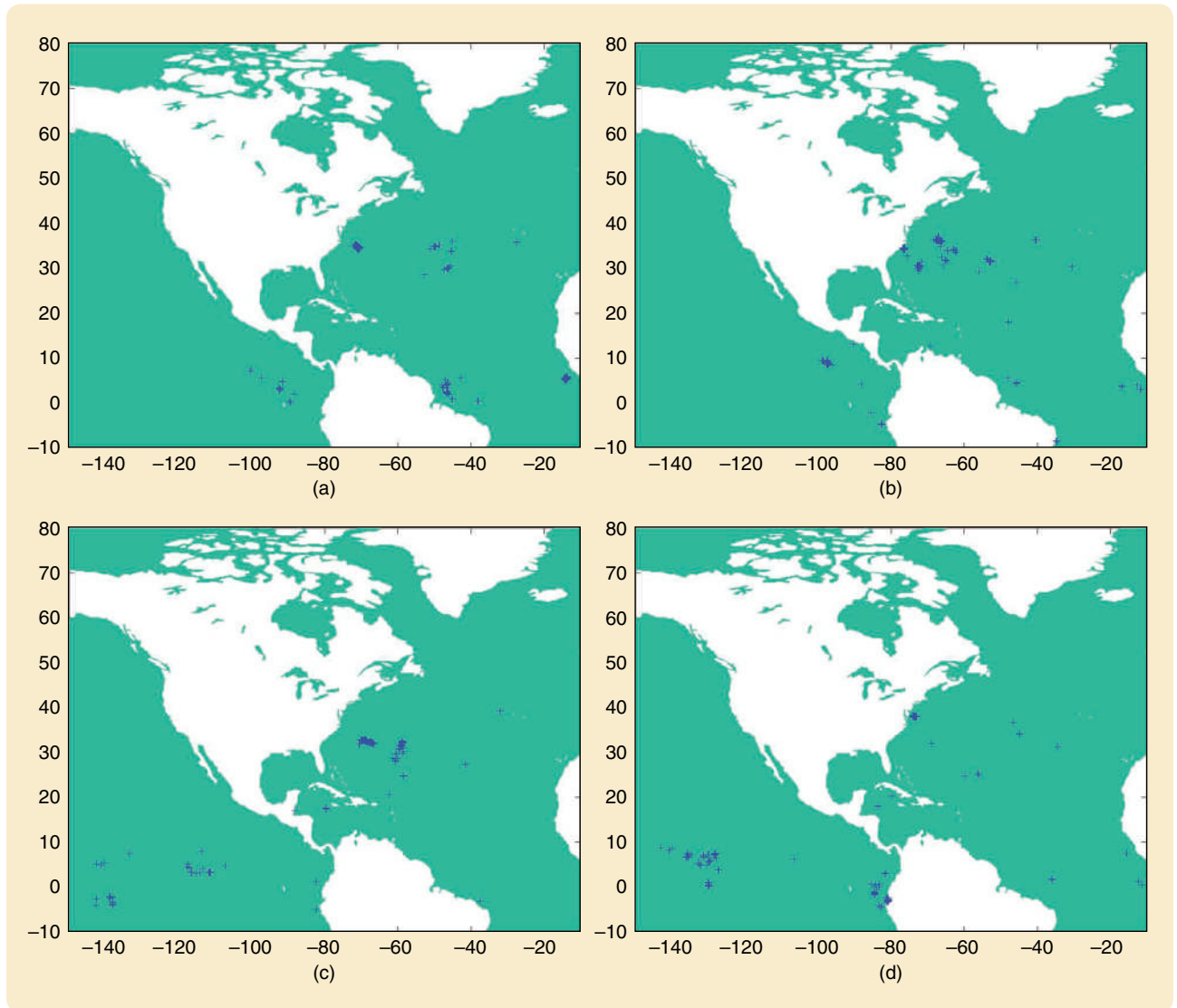


FIGURE 14 Locations of weather anomaly obtained based on the error between the actual temperature and the prediction of the autonomous kernel observer. The landmass is shown in white, and the ocean is in green. Marked locations have an error greater than two standard deviations above the mean error. (a) May 6, 2012. (b) May 7, 2012. (c) May 28, 2012. (d) May 29, 2012.

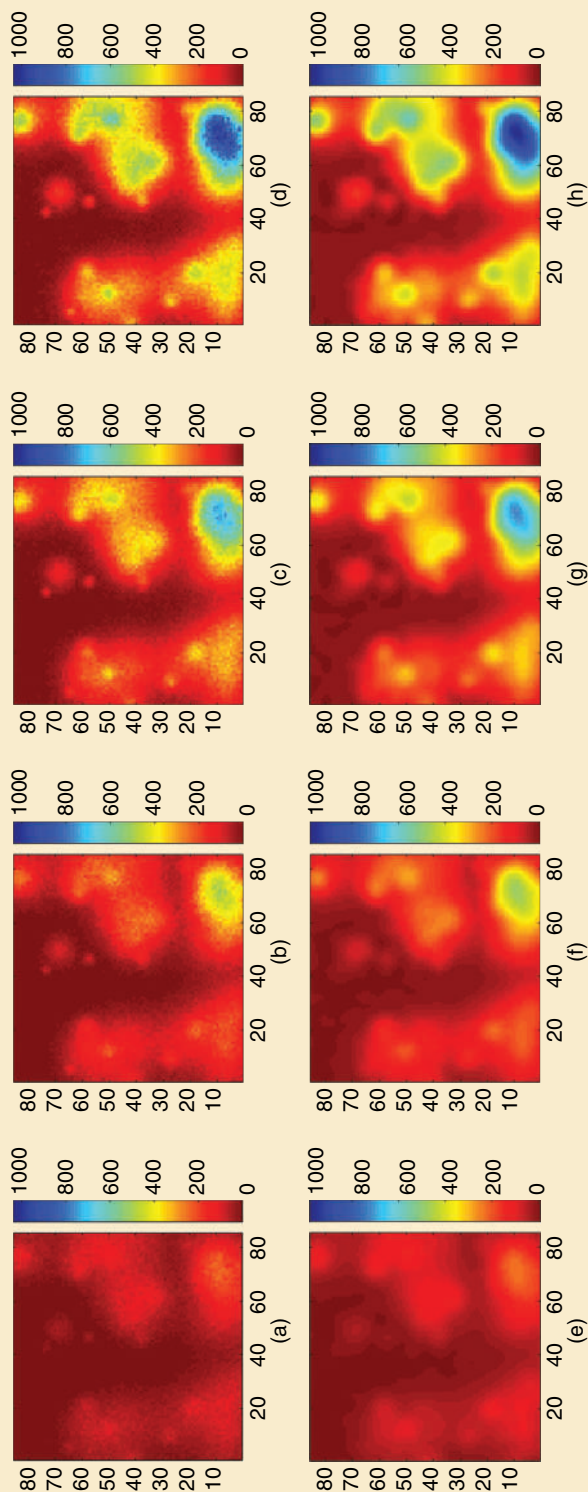


FIGURE 15 A visualization of weed density growth during 20 days. (a)–(d) The original visualization. (e)–(h) The evolving Gaussian process. (a) Snapshot 2. (b) Snapshot 28. (c) Snapshot 54. (d) Snapshot 80. (e) Snapshot 2. (f) Snapshot 28. (g) Snapshot 54. (h) Snapshot 80.

elapse before weeding starts. The number of emerging weeds in each cell N_{emerge} is a randomly generated Poisson variable with mean $\lambda(x, y, t)$. The weed density in each cell $\zeta(x, y, t)$ rises as seeds emerge from the bank. The maximum weed height at each cell $\delta(x, y, t)$ increases from zero at a fixed rate of Γ in/day.

The main challenge with weeding robots is that they have access only to sparse data about plant density and height and no information about the seed bank. In fact, the robots can sense only the row they are in and potentially the adjacent rows on either side. They have no information about locations in the field that have not been visited. Hence, they must try to predict the global weed density given sparse information. This is a great application for the kernel observer techniques studied in this article. To demonstrate the proof of concept, a kernel observer model was trained using the weed density data generated by our simulations so that robots in a field can quickly globally infer the state of the field from partial measurements based on a few rows. RBF kernels were used, with a bandwidth that was approximately five cells (4.5 m) wide. These were centered throughout the domain by using the algorithm in [70].

After training a GP on each snapshot of the data, a weight vector trajectory was obtained. Linear least squares was used to determine the best linear transition in the weight space for this trajectory. Images were predicted by feeding forward the initial condition through this linear transition as a comparison with the original data. The model was able to approximate the weed density growth data very well, with the percent error averaging 5%. These results show the feasibility of modeling weed growth using E-GPs. Ongoing work involves a massive field campaign to collect the weed density data required to teach these models. Such data sets are currently not available, and the TerraSentia robots discussed in “Key Control Problems in Agriculture” are being used to collect this data.

Comparison of Eigenvalues and Koopman Modes With Dynamic Mode Decomposition

To empirically verify our theoretical results, the eigenvalues and Koopman modes derived from $\hat{A}$ in this E-GP model are compared with the eigenvalues and Koopman modes derived by the well-known DMD algorithm, using data from fluid flowing past a cylinder at different Reynolds numbers. The results are presented in Figures 16 and 17. As shown, the modes generated by the E-GP match those generated by DMD at least as well as the E-GP predictions match the original data.

CONCLUSION

Machine learning techniques such as GP regression and deep learning are providing increasingly more powerful ways of learning predictive models. Since these techniques are limited by the data sets that they are trained with, their predictions are not always reliable when forecasting real-world, complex spatiotemporally evolving phenomena that have high variability. Years of weather data cannot be a reliable predictor of the current weather, nor can data from one farm directly relate to another. Because of this, when designing complex cyberphysical systems, such as teams of mechanically weeding robots, engineers have devised ways to supplement machine learning model predictions with real measurements. This is not always immediate when one works in the abstract feature spaces utilized in machine learning models.

This tutorial presented kernel observers and their extension, E-GPs, which are formed by embedding a linear dynamical system in the RKHSs generated by the GP model. The parameters of the linear dynamical system can be trained to approximately predict the evolution. This leads to powerful and generalizable models, as demonstrated by results for predicting solutions to the Navier–Stokes equations. When supplemented with a predict-and-correct

TABLE 2 Seed bank density parameters.

Parameter	Gini Coefficient of Concentration	S_0 (Seeds/Cell)	Number of Patches	Patch Size (Cells in X and Y)
Range	[0.31,0.35]	[600,1560]	50	[0,20]

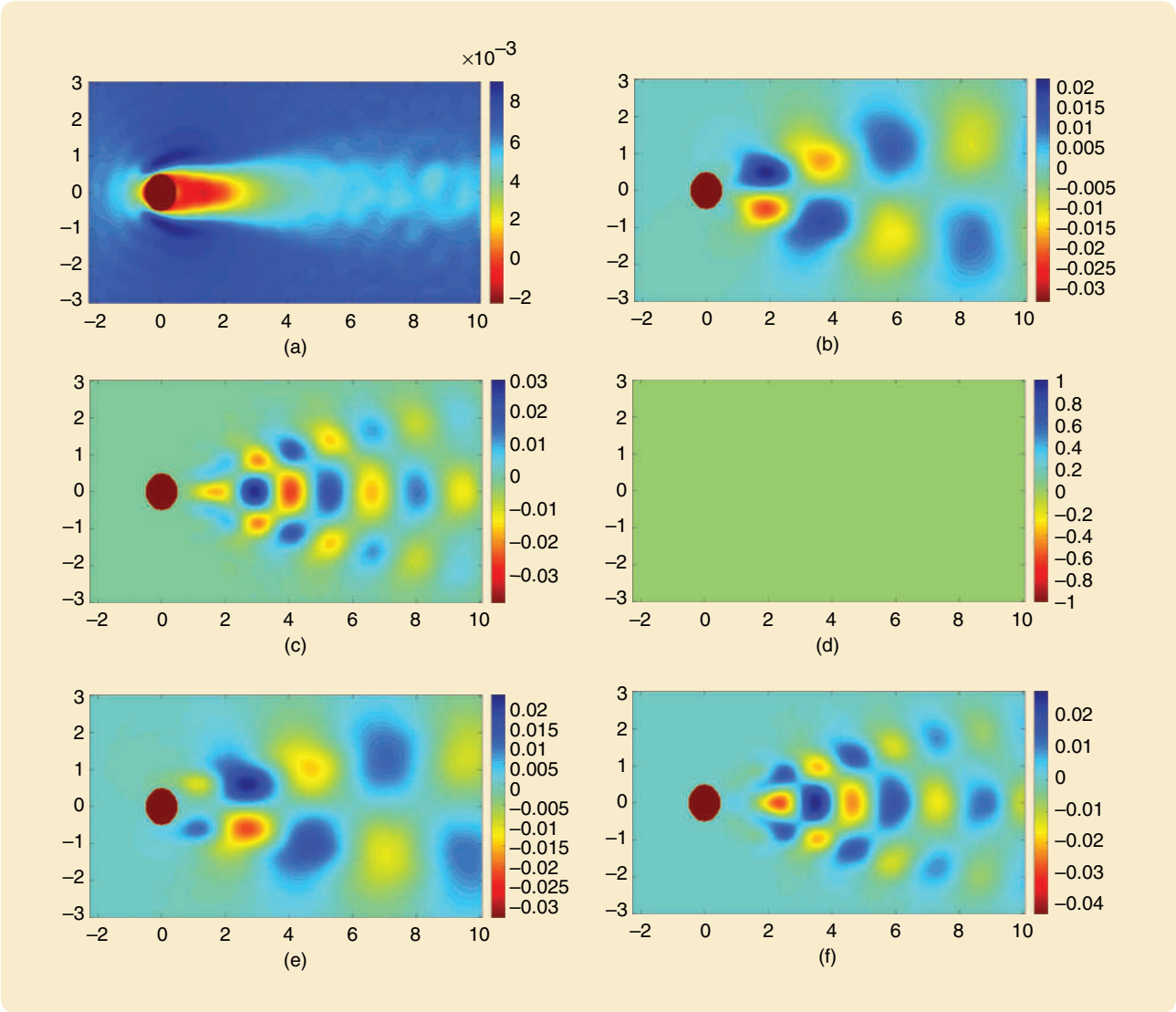


FIGURE 16 The primary Koopman modes of a flow past a cylinder at Reynolds number 100. (a)–(f) The evolving Gaussian process. (g)–(l) The dynamic mode decomposition. (a) 0°, real component. (b) 6°, real component. (c) 12°, real component. (d) 0°, imaginary component. (e) 6°, imaginary component. (f) 12°, imaginary component. (Continued)